

Chapter 3

Simple Lie algebras. Classification and representations. Roots and weights

3.1 Cartan subalgebra. Roots. Canonical form of the algebra

We consider a semi-simple (*i.e.* with no abelian ideal) Lie algebra of finite dimension. We want to construct a canonical form of commutation relations modeled on the case of $SU(2)$

$$[J_z, J_{\pm}] = \pm J_{\pm} \quad [J_+, J_-] = 2J_z . \quad (3.1)$$

It will be important to consider the algebra over $\mathbb{C}$, at the price of “complexifying” it if it was originally real. The adjoint representation will be used. As it is a faithful representation for a semi-simple algebra, (*i.e.* $\text{ad } X = 0 \Rightarrow X = 0$, see exercise B of Chap. 2), no information is lost.

It may also be useful to remember that the complex algebra has a real compact version, in which the real structure constants lead to a negative definite Killing form, and, as the representations can be taken unitary, the elements of the Lie algebra (the infinitesimal generators) may be taken as Hermitian (or antiHermitian, depending on our conventions).

3.1.1 Cartan subalgebra

We define first the notion of *Cartan subalgebra*. This is a *maximal abelian subalgebra* of $\mathfrak{g}$ such that all its elements are diagonalisable (hence simultaneously diagonalisable) in the adjoint representation. That such an algebra exists is non trivial and must be established, but we shall admit it.

If we choose to work with the unitary form of the adjoint representation, the elements of $\mathfrak{g}$ are Hermitian matrices, and assuming that the elements of $\mathfrak{h}$ are commuting among themselves ensures that they are simultaneously diagonalizable.

This Cartan subalgebra is non unique, but one may prove that two distinct choices are related by an automorphism of the algebra $\mathfrak{g}$.

For instance if $\mathfrak{g}$ is the Lie algebra of a Lie group G and if $\mathfrak{h}$ is a Cartan subalgebra of $\mathfrak{g}$, any conjugate ghg^{-1} of $\mathfrak{h}$ by an arbitrary element of G is another Cartan subalgebra.

Let $\mathfrak{h}$ be a Cartan subalgebra, call ℓ its dimension, it is independent of the choice of $\mathfrak{h}$ and it is called the *rank* of $\mathfrak{g}$. For $\mathfrak{su}(2)$, this rank is 1, (the choice of J_z for example); for $\mathfrak{su}(n)$, the rank is $n - 1$. Indeed for $\mathfrak{su}(n)$, a Cartan algebra is generated¹ by diagonal traceless matrices, a basis of which is given by the $n - 1$ matrices

$$H_1 = \text{diag}(1, -1, 0, \dots, 0), H_2 = \text{diag}(0, 1, -1, 0, \dots, 0), \dots, H_{n-1} = \text{diag}(0, \dots, 0, 1, -1). \quad (3.2)$$

An arbitrary matrix of the Lie algebra, (in that representation), (anti-)Hermitian and traceless, is diagonalisable by a unitary transformation; its diagonal form is traceless and is thus expressed as a linear combination of the h_j ; the original matrix is thus conjugate by a unitary transformation of a linear combination of the h_j . This is a general property, and one proves (Cartan, see [Bu], chap. 16) that

If $\mathfrak{g}$ is the Lie algebra of a group G , any element of $\mathfrak{g}$ is conjugate by G of an element of $\mathfrak{h}$.

Application. Canonical form of antisymmetric matrices. Using the previous statement, prove the

Proposition *If $A = A^* = -A^T$ is a real skew-symmetric matrix of dimension N , one may find a real orthogonal matrix O such that $A = ODO^T$ where $D = \text{diag}\left(\begin{pmatrix} 0 & \mu_j \\ -\mu_j & 0 \end{pmatrix}\right)_{j=1, \dots, n}$ if $N = 2n$ and $D = \text{diag}\left(0, \begin{pmatrix} 0 & \mu_j \\ -\mu_j & 0 \end{pmatrix}\right)_{j=1, \dots, n}$ if $N = 2n + 1$, with real μ_j .*

If one allows the complexification of orthogonal matrices, one may fully diagonalise the matrix A in the form $D = \text{diag}\left(\begin{pmatrix} i\mu_j & 0 \\ 0 & -i\mu_j \end{pmatrix}\right)_{j=1, \dots, n}$ or $D = \text{diag}\left(0, \begin{pmatrix} i\mu_j & 0 \\ 0 & -i\mu_j \end{pmatrix}\right)_{j=1, \dots, n}$. For a proof making only use of matrix theory, see for example [M.L. Mehta, *Elements of Matrix Theory*, p 41].

3.1.2 Canonical basis of the Lie algebra

Let $H_i, i = 1, \dots, \ell$ be a basis of $\mathfrak{h}$. It is convenient to choose the H_i Hermitian. By definition $[H_i, H_j] = 0$, (abelian subalgebra) or more precisely, since we are in the adjoint representation,

$$[\text{ad } H_i, \text{ad } H_j] = 0. \quad (3.3)$$

We may thus diagonalise simultaneously these $\text{ad } H_i$. We already know (some?) eigenvectors of vanishing eigenvalue since $\forall i, j, \text{ad } H_i H_j = 0$, and we may complete them to make a basis by finding a set of eigenvectors E_α linearly independent of the H_j

$$\text{ad } H_i E_\alpha = \alpha_{(i)} E_\alpha \quad (3.4)$$

i.e. a set of elements of $\mathfrak{g}$ such that

$$[H_i, E_\alpha] = \alpha_{(i)} E_\alpha, \quad (3.5)$$

¹We use momentarily the “representation of definition” (made of $n \times n$ matrices) rather than the adjoint representation.

with the $\alpha_{(i)}$ not all vanishing (otherwise the subalgebra $\mathfrak{h}$ would not be maximal).

The space $\mathfrak{h}^$.* In these expressions, the $\alpha_{(i)}$ are eigenvalues of the operators $\text{ad } H_i$. Since we chose Hermitian $\text{ad } H_i$, their eigenvalues $\alpha_{(i)}$ are real. By linearity, for an arbitrary element of $\mathfrak{h}$ written as $H = \sum_i h^i H_i$,

$$\text{ad}(H)E_\alpha = \alpha(H)E_\alpha, \quad (3.6)$$

and the eigenvalue of $\text{ad}(H)$ on E_α is $\alpha(H) := \sum_i h^i \alpha_{(i)}$, which is a *linear form* on $\mathfrak{h}$. In general linear forms on a vector space E form a vector space E^* , called the dual space of E . One may thus consider the *root* α , of components $\alpha_{(i)}$, as a vector of the dual space of $\mathfrak{h}$, hence $\alpha \in \mathfrak{h}^*$, the *root space*. Note that $\alpha(H_i) = \alpha_{(i)}$.

Roots enjoy the following properties (*)

1. if α is a root, $-\alpha$ in another root;
2. the eigenspace of the eigenvalue α is of dimension 1 (no multiplicity);
3. if α is a root, the only roots of the form $\lambda\alpha$ are $\pm\alpha$;
4. roots α generate all the dual space $\mathfrak{h}^*$.

For proofs of 1., 2., 3., see below, for 4. see exercise A.

Number of roots. Since the H_j are diagonalisable, the total number of their eigenvectors E_α and H_i must be equal to the dimension of the space, here the dimension d of the adjoint representation, *i.e.* of the Lie algebra $\mathfrak{g}$. As any (non vanishing by definition) root comes along with its opposite, the number of roots α is even and equal to $d - \ell$ (with $\ell = \text{rank}(\mathfrak{g})$). We denote Δ the set of roots.

In the basis $\{H_i, E_\alpha\}$ of $\mathfrak{g}$, the Killing form takes a simple form

$$(H_i, E_\alpha) = 0 \quad (E_\alpha, E_\beta) = 0 \quad \text{unless } \alpha + \beta = 0. \quad (3.7)$$

To show that, we write $(H, [H', E_\alpha]) = \alpha(H')(H, E_\alpha)$, and also, using the definition of the Killing form and the cyclicity of the trace

$$(H, [H', E_\alpha]) = \text{tr}(\text{ad } H[\text{ad } H', \text{ad } E_\alpha]) = \text{tr}([\text{ad } H, \text{ad } H'] \text{ad } E_\alpha) = 0 \quad (3.8)$$

since $[\text{ad } H, \text{ad } H'] = 0$. It follows that $\forall H, H' \in \mathfrak{h}$, $\alpha(H')(H, E_\alpha) = 0$, hence that $(H, E_\alpha) = 0$. Likewise

$$([H, E_\alpha], E_\beta) = \alpha(H)(E_\alpha, E_\beta) = -(E_\alpha, [H, E_\beta]) = -\beta(H)(E_\alpha, E_\beta) \quad (3.9)$$

again by the cyclicity of the trace, and thus $(E_\alpha, E_\beta) = 0$ if $\exists H : (\alpha + \beta)(H) \neq 0$, *i.e.* if $\alpha + \beta \neq 0$. Note that the point 1. in (*) above follows simply from (3.7): if $-\alpha$ were not a root, E_α would be orthogonal to all elements of the basis hence to any element of $\mathfrak{g}$, and the form would be degenerate, contrary to the hypothesis of semi-simplicity (and Cartan's criterion). For an elegant proof of points 2. et 3. of (*), see [OR, p. 29].

The restriction of this form to the Cartan subalgebra is non-degenerate, since otherwise one would have $\exists H \in \mathfrak{h}$, $\forall H' \in \mathfrak{h} : (H, H') = 0$, but $(H, E_\alpha) = 0$, thus $\forall X \in \mathfrak{g}$, $(H, X) = 0$ and the form would be degenerate, contrary to the hypothesis of semi-simplicity (and Cartan's criterion, Chap. 1, §4.4). The Killing form being non-degenerate on $\mathfrak{h}$, it induces an isomorphism between $\mathfrak{h}$ and $\mathfrak{h}^*$: to $\alpha \in \mathfrak{h}^*$ one associates the unique $H_\alpha \in \mathfrak{h}$ such that

$$\forall H \in \mathfrak{h} \quad (H_\alpha, H) := \alpha(H). \quad (3.10)$$

(Or said differently, one solves the linear system $g_{ij}h_\alpha^j = \alpha_{(i)}$ which is of Cramer type since $g_{ij} = (H_i, H_j)$ is invertible.) One has also a bilinear form on $\mathfrak{h}^*$ inherited from the Killing form

$$\langle \alpha, \beta \rangle := (H_\alpha, H_\beta) , \quad (3.11)$$

which we are going to use in § 2 to study the geometry of the root system.

It remains to find the commutation relations of the E_α among themselves. Using the Jacobi identity, one finds that

$$\text{ad } H_i[E_\alpha, E_\beta] = [H_i, [E_\alpha, E_\beta]] = [E_\alpha, [H_i, E_\beta]] - [E_\beta, [H_i, E_\alpha]] = (\alpha + \beta)_{(i)}[E_\alpha, E_\beta] . \quad (3.12)$$

Invoking the trivial multiplicity (=1) of roots, one sees that three cases may occur. If $\alpha + \beta$ is a root, $[E_\alpha, E_\beta]$ is proportional to $E_{\alpha+\beta}$, with a proportionality coefficient $N_{\alpha\beta}$ which will be shown below to be non zero (see § 3.2.1 and exercise B). If $\alpha + \beta \neq 0$ is not a root, $[E_\alpha, E_\beta]$ must vanish. Finally if $\alpha + \beta = 0$, $[E_\alpha, E_{-\alpha}]$ is an eigenvector of all $\text{ad } H_i$ with a vanishing eigenvalue, thus $[E_\alpha, E_{-\alpha}] = H \in \mathfrak{h}$. To determine that H , let us proceed like in (3.9)

$$\begin{aligned} (H_i, [E_\alpha, E_{-\alpha}]) &= \text{tr}(\text{ad } H_i [\text{ad } E_\alpha, \text{ad } E_{-\alpha}]) = \text{tr}([\text{ad } H_i, \text{ad } E_\alpha] \text{ad } E_{-\alpha}) \\ &= \alpha_{(i)}(E_\alpha, E_{-\alpha}) = (H_i, H_\alpha)(E_\alpha, E_{-\alpha}) \end{aligned} \quad (3.13)$$

hence

$$[E_\alpha, E_{-\alpha}] = (E_\alpha, E_{-\alpha})H_\alpha . \quad (3.14)$$

To recapitulate, we have constructed a canonical basis of the algebra $\mathfrak{g}$

$$\begin{aligned} [H_i, H_j] &= 0 \\ [H_i, E_\alpha] &= \alpha_{(i)}E_\alpha \\ [E_\alpha, E_\beta] &= \begin{cases} N_{\alpha\beta}E_{\alpha+\beta} & \text{if } \alpha + \beta \text{ is a root} \\ (E_\alpha, E_{-\alpha})H_\alpha & \text{if } \alpha + \beta = 0 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

(3.15)

Up to that point, the normalisation of the vectors H_i and E_α has not been fixed. It is common to choose, in accord with (3.7)

$$(H_i, H_j) = \delta_{ij} \quad (E_\alpha, E_\beta) = \delta_{\alpha+\beta, 0} .$$

(3.15)

(Indeed, the restriction of the Killing form to $\mathfrak{h}$, after multiplication by i to make the $\text{ad } H_i$ Hermitian, is *positive* definite.) With that normalisation, H_α defined above by (3.10) satisfies also

$$H_\alpha = \alpha.H := \alpha_{(i)}H_i . \quad (3.16)$$

Note that E_α , $E_{-\alpha}$ and H_α form an $\mathfrak{su}(2)$ subalgebra

$$[H_\alpha, E_{\pm\alpha}] = \pm \langle \alpha, \alpha \rangle E_{\pm\alpha} \quad [E_\alpha, E_{-\alpha}] = H_\alpha . \quad (3.17)$$

(This is in fact $H_\alpha / \langle \alpha, \alpha \rangle$ that we identify with J_z , and that observation will be used soon.) Any semi-simple algebra thus contains an $\mathfrak{su}(2)$ algebra associated to each of its roots.

Note that with the normalisations of (3.15), the Killing metric reads in the basis $\{H_i, E_\alpha, E_{-\alpha}\}$

$$g_{ab} = \begin{pmatrix} \mathbb{I}_\ell & & & 0 \\ & 0 & 1 \\ & 1 & 0 \\ 0 & & \ddots & \\ & & & 0 & 1 \\ & & & 1 & 0 \end{pmatrix} \quad (3.18)$$

where the first block is an identity matrix of dimension $\ell \times \ell$.

3.2 Geometry of root systems

3.2.1 Scalar products of roots. The Cartan matrix

As noticed in (3.11), the space of roots, *i.e.* the space (of dimension ℓ , see point 4. in (*) above) generated by the $d - \ell$ roots α inherits the Euclidean metric of $\mathfrak{h}$

$$\langle \alpha, \beta \rangle := (H_\alpha, H_\beta) = \alpha(H_\beta) = \beta(H_\alpha) = (\alpha.H, \beta.H) = \sum_i \alpha_{(i)} \beta_{(i)} , \quad (3.19)$$

where the various expressions aim at making the reader familiar with the notations introduced above. (Only the last two expressions depend on the choice of normalisation (3.15).) We shall now show that the geometry –lengths and angles– of roots is strongly constrained. First it is good to remember the lessons of the $\mathfrak{su}(2)$ algebra: in a representation of finite dimension, J_z has integer or half-integer eigenvalues. Thus here, where each $\frac{H_\alpha}{\langle \alpha, \alpha \rangle}$ plays the role of a J_z and has E_β as eigenvectors, $\text{ad } H_\alpha E_\beta = \langle \alpha, \beta \rangle E_\beta$, *i.e.*

$$[H_\alpha, E_\beta] = \langle \alpha, \beta \rangle E_\beta \quad (3.20)$$

we may conclude that

$$2 \frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle} = m \in \mathbb{Z} . \quad (3.21)$$

Root chains

It is in fact useful to refine the previous discussion. Like in the case of $\mathfrak{su}(2)$, the idea is to repeatedly apply the “raising” E_α and “lowering” $E_{-\alpha}$ operators (aka *ladder operators*) on a given eigenvector E_β . We saw that if α and β are two distinct roots, with $\alpha + \beta \neq 0$, it may happen that $\beta \pm \alpha$ are also roots. Let $p \leq 0$ be the smallest integer such that $(\text{ad } E_{-\alpha})^{|p|} E_\beta$ is non

zero, *i.e.* that $\beta + p\alpha$ is a root, and let $q \geq 0$ be the largest integer such that $(\text{ad } E_\alpha)^q E_\beta$ is non zero, *i.e.* that $\beta + q\alpha$ is a root. We call the subset of roots $\{\beta + p\alpha, \beta + (p+1)\alpha, \dots, \beta, \dots, \beta + q\alpha\}$ the α -chain through β . Note that the $E_{\beta'}$, when β' runs along that chain, form a basis of a finite dimensional representation of the $\text{su}(2)$ algebra generated by H_α and $E_{\pm\alpha}$. According to what we know about these representations of $\text{su}(2)$, the lowest and highest eigenvalues of H_α are opposite

$$\langle \alpha, \beta + p\alpha \rangle = -\langle \alpha, \beta + q\alpha \rangle$$

or $2\langle \beta, \alpha \rangle = -(q+p)\langle \alpha, \alpha \rangle$, thus with the notation (3.21)

$$m = -p - q. \quad (3.22)$$

This construction also shows that $\beta - m\alpha = \beta + (p+q)\alpha$ is in the α -chain through β , (since $p \leq -m \leq q$), hence that this is a root.

Remark. The discussion of § 3.1 left the coefficients $N_{\alpha\beta}$ undetermined. One shows (see Exercise B), using the commutation relations of the E 's along a chain that the coefficients $N_{\alpha\beta}$ satisfy non linear relations and that they are determined up to signs by the geometry of the root system according to

$$|N_{\alpha\beta}| = \sqrt{\frac{1}{2}(1-p)q\langle \alpha, \alpha \rangle}. \quad (3.23)$$

Note that, as stated before, $N_{\alpha\beta}$ vanishes only if $q = 0$, *i.e.* if $\alpha + \beta$ is not a root.

Weyl group

For any vector x in the root space $\mathfrak{h}^*$, define the linear transformation

$$w_\alpha(x) = x - 2 \frac{\langle \alpha, x \rangle}{\langle \alpha, \alpha \rangle} \alpha. \quad (3.24)$$

This is a reflection in the hyperplane orthogonal to α : $(w_\alpha)^2 = I$, $w_\alpha(\alpha) = -\alpha$, and $w_\alpha(x) = x$ if x is orthogonal to α . This is of course an isometry, since it preserves the scalar product: $\langle w_\alpha(x), w_\alpha(y) \rangle = \langle x, y \rangle$. Such a w_α is called a Weyl reflection. By definition the *Weyl group* W is the group generated by the w_α , *i.e.* the set of all possible products of w_α over roots α . Thanks to the remark following (3.22), if α and β are two roots, $w_\alpha(\beta) = \beta - m\alpha$ is also a root. The set of roots is thus globally invariant under the action of the Weyl group. The group W is completely determined by its action on roots, which is a permutation. W is thus a subgroup of the permutation group of the finite set Δ , hence a finite group².

Example : for the algebra $\text{su}(n)$, one finds that $W = \mathcal{S}_n$, the permutation group of n objects, see below in § 3.3.2.

Note that if $\beta_+ = \beta + q\alpha$ is the highest root in the α -chain through β , and $\beta_- = \beta + p\alpha$ the lowest one, $w_\alpha(\beta_\pm) = \beta_\mp$ and more generally, the roots of the chain are swapped pairwise under the action of w_α . The chain is thus invariant by w_α . (This is a generalisation of the $m \leftrightarrow -m$ symmetry of the $\text{su}(2)$ “multiplets” $(-j, -j+1, \dots, j-1, j)$.)

²This property is far from trivial: generically, when m vectors are given in the Euclidean space $\mathbb{R}^m$, the group generated by reflections in the hyperplanes orthogonal to these vectors is infinite. You need very peculiar configurations of vectors to make the group finite. Finite reflection groups have been classified by Coxeter. Weyl groups of simple algebras form a subset of Coxeter groups.

Positive roots, simple roots. Cartan matrix

Roots are not linearly independent in $\mathfrak{h}^*$. One may show that one can partition their set Δ into “positive” and “negative” roots, the opposite of a positive root being negative, and find a basis α_i , $i = 1, \dots, \ell$ of ℓ *simple* roots, such that any positive (resp. negative) root is a linear combination with non negative (resp non positive) integer coefficients of these simple roots. As a consequence, a simple root cannot be written as the sum of two positive roots (check !).

Neither the choice of a set of positive roots, nor that of a basis of simple roots is unique. One goes from a basis of simple roots to another one by some operation of the Weyl group.

If α and β are simple roots, $\alpha - \beta$ cannot be a root (why?). The integer p in the previous discussion thus vanishes and $m = -q \leq 0$. It follows that $\langle \alpha, \beta \rangle \leq 0$.

The scalar product of two simple roots is non positive. (3.25)

We now define the Cartan matrix

$$C_{ij} = 2 \frac{\langle \alpha_i, \alpha_j \rangle}{\langle \alpha_j, \alpha_j \rangle}. \quad (3.26)$$

Beware, that matrix is *a priori* non symmetric.³ Its diagonal elements are 2, its off-diagonal elements are ≤ 0 integers.

One must remember that the scalar product appearing in the numerator of (3.26) is positive definite. According to the Schwarz inequality, $\langle \alpha, \beta \rangle^2 \leq \langle \alpha, \alpha \rangle \langle \beta, \beta \rangle$ with equality only if α and β are colinear. This property, together with the integrity properties of their elements, suffices to classify all possible Cartan matrices, as we shall now see.

Write $\langle \alpha_i, \alpha_j \rangle = \|\alpha_i\| \|\alpha_j\| \cos \widehat{\alpha_i, \alpha_j}$. Then by multiplying or dividing the two equations (3.21) for the pair $\{\alpha_i, \alpha_j\}$, $i \neq j$, namely $C_{ij} = m_i \leq 0$ and $C_{ji} = m_j \leq 0$, where the property (3.25) above has been taken into account, one finds that if $i \neq j$,

$$\left. \begin{aligned} \cos \widehat{\alpha_i, \alpha_j} &= -\frac{1}{2} \sqrt{m_i m_j} \\ \frac{\|\alpha_i\|}{\|\alpha_j\|} &= \sqrt{\frac{m_i}{m_j}} \end{aligned} \right\} \quad \text{with } m_i, m_j \in \mathbb{N}, \quad (3.27)$$

and the value -1 of the cosinus is impossible, since $\alpha_i \neq -\alpha_j$ by assumption, so that the only possible values of that cosinus are $0, -\frac{1}{2}, -\frac{\sqrt{2}}{2}, -\frac{\sqrt{3}}{2}$, *i.e.* the only possible angles between simple roots are $\frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}$ or $\frac{5\pi}{6}$, with ratios of lengths of roots respectively equal to ?(undetermined), $1, \sqrt{2}, \sqrt{3}$.

There exists of course only one algebra of rank 1, viz the (complexified) $\mathfrak{su}(2)$ algebra, (3.1) or (3.17). It will be called A_1 below. It is then easy to classify the possible algebras of rank 2. The four cases are depicted on Fig. 3.1, with their Cartan matrices reading

$$A_2 : \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \quad B_2 : \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix} \quad G_2 : \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix} \quad D_2 : \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}. \quad (3.28)$$

The nomenclature, A_2 , B_2 , G_2 and D_2 , is conventional, and so is the numbering of roots. The latter case, D_2 , which has $\langle \alpha_1, \alpha_2 \rangle = 0$, is mentioned here for completeness: it corresponds to

³Also, beware that some authors call Cartan matrix the transpose of (3.26)!

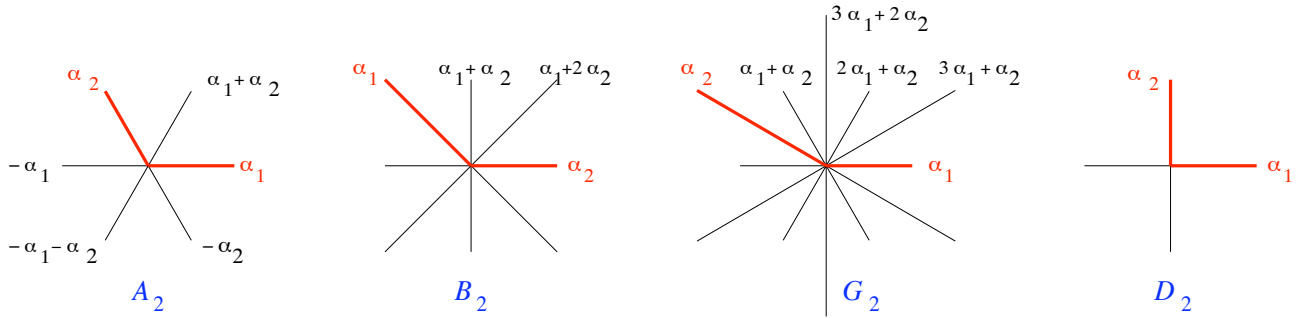


Figure 3.1: Root systems of rank 2. The two simple roots are drawn in thick lines. For the algebras B_2 , G_2 et D_2 , only positive roots have been labelled.

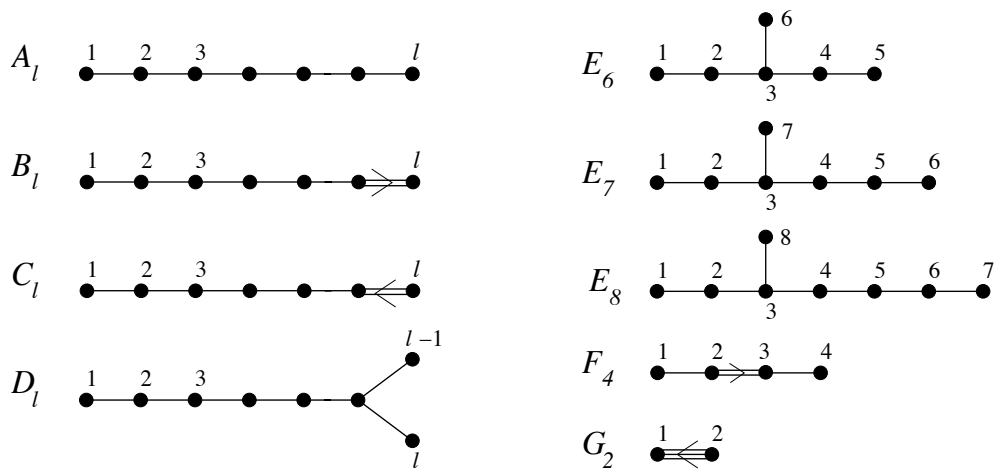


Figure 3.2: Dynkin diagrams

a semi-simple algebra, the direct sum of two A_1 algebras. (Nothing forces its two roots to be of equal length.)

In general, if the set of roots may be split into two mutually orthogonal subsets, one sees that the Lie algebra decomposes into a direct sum of two algebras, and vice versa. Recalling that any semi-simple algebra may be decomposed into the direct sum of simple subalgebras (see end of Chap. 1), in the following we consider only simple algebras.

Dynkin diagram

For higher rank, *i.e.* for higher dimension of the root space, it becomes difficult to visualise the root system. Another representation is adopted, by encoding the Cartan matrix into a diagram in the following way: with each simple root is associated a vertex of the diagram; two vertices are linked by an edge iff $\langle \alpha_i, \alpha_j \rangle \neq 0$; the edge is simple if $C_{ij} = C_{ji} = -1$ (angle of $2\pi/3$, equal lengths); it is double (resp. triple) if $C_{ij} = -2$ (resp. -3) and $C_{ji} = -1$ (angle of $\frac{3\pi}{4}$ resp. $\frac{5\pi}{6}$, with a length ratio of $\sqrt{2}$, resp. $\sqrt{3}$) and then carries an arrow (or rather a sign $>$) from i to j indicating which root is the longest. (Beware that some authors use the opposite convention for arrows!).

3.2.2 Root systems of simple algebras. Cartan classification

The analysis of all possible cases led Cartan⁴ to a classification of simple complex Lie algebras, in terms of four infinite families and five exceptional cases. The traditional notation is the following

$$A_\ell, \quad B_\ell, \quad C_\ell, \quad D_\ell, \quad E_6, \quad E_7, \quad E_8, \quad F_4, \quad G_2. \quad (3.29)$$

In each case, the lower index gives the rank of the algebra. The geometry of the root system is encoded in the Dynkin diagrams of Fig. 3.2.

The four infinite families are identified with the (complexified) Lie algebras of classical groups

$$A_\ell = \mathfrak{sl}(\ell + 1, \mathbb{C}), \quad B_\ell = \mathfrak{so}(2\ell + 1, \mathbb{C}), \quad C_\ell = \mathfrak{sp}(2\ell, \mathbb{C}), \quad D_\ell = \mathfrak{so}(2\ell, \mathbb{C}). \quad (3.30)$$

or with their unique compact real form, respectively $A_\ell = \mathfrak{su}(\ell + 1)$, $B_\ell = \mathfrak{so}(2\ell + 1)$, $C_\ell = \mathfrak{usp}(\ell)$, $D_\ell = \mathfrak{so}(2\ell)$.

The “exceptional algebras” $E_6, \dots, G_2$ have respective dimensions 78, 133, 248, 52 and 14. Those are algebras of ... exceptional Lie groups ! The *group* G_2 is the group of automorphisms of octonions, F_4 is itself an automorphism group of octonion matrices, etc.

Among these algebras, the algebras A, D, E , whose roots have the same length, are called *simply laced*. A curious observation is that many problems, finite subgroups of $\mathfrak{su}(2)$, “simple” singularities, “minimal conformal field theories”, etc, are classified by the same *ADE* scheme... but this is another story!

The real forms of these simple complex algebras have also been classified by Cartan. One finds 12 infinite series and 23 exceptional cases!

3.2.3 Chevalley basis

There exists another basis of the Lie algebra $\mathfrak{g}$, called Chevalley basis, with brackets depending only on the Cartan matrix. Let h_i, e_i et $f_i, i = 1, \dots, \ell$, be generators attached to simple roots α_i according to

$$e_i = \left(\frac{2}{\langle \alpha_i, \alpha_i \rangle} \right)^{\frac{1}{2}} E_{\alpha_i}, \quad f_i = \left(\frac{2}{\langle \alpha_i, \alpha_i \rangle} \right)^{\frac{1}{2}} E_{-\alpha_i}, \quad h_i = \frac{2\alpha_i \cdot H}{\langle \alpha_i, \alpha_i \rangle}. \quad (3.31)$$

Their commutation relations read

$$\begin{aligned} [h_i, h_j] &= 0 \\ [h_i, e_j] &= C_{ji} e_j \\ [h_i, f_j] &= -C_{ji} f_j \\ [e_i, f_j] &= \delta_{ij} h_j \end{aligned} \quad (3.32)$$

(check!). The algebra is generated by the e_i, f_i, h_i and all their commutators, constrained by (3.32) and by the “Serre relations”

$$\begin{aligned} \text{ad}(e_i)^{1-C_{ji}} e_j &= 0 \\ \text{ad}(f_i)^{1-C_{ji}} f_j &= 0. \end{aligned} \quad (3.33)$$

This proves that the whole algebra is indeed encoded in the data of the simple roots and of their geometry (Cartan matrix or Dynkin diagram).

Note also the remarkable and *a priori* not obvious property that in that basis, all the structure constants (coefficients of the commutation relations) are *integers*.

⁴This classification work, undertaken by Killing, was corrected and completed by É. Cartan, and later simplified by van der Waerden, Dynkin, ...

3.2.4 Coroots. Highest root. Coxeter number and exponents

We give here some complements on notations and concepts that are encountered in the study of simple Lie algebras and of their root systems.

As the combination

$$\alpha_i^\vee := \frac{2}{\langle \alpha_i, \alpha_i \rangle} \alpha_i, \quad (3.34)$$

for α_i a simple root, appears frequently, it is given the name of *coroot*. The Cartan matrix may be rewritten as

$$C_{ij} = \langle \alpha_i, \alpha_j^\vee \rangle. \quad (3.35)$$

The *highest root* θ is the positive root with the property that the sum of its components in a basis of simple roots is maximal: one proves that this characterizes it uniquely. Its components in the basis of simple roots and in that of coroots

$$\theta = \sum_i a_i \alpha_i, \quad \frac{2}{\langle \theta, \theta \rangle} \theta = \sum_i a_i^\vee \alpha_i^\vee, \quad (3.36)$$

called *Kac labels*, *resp dual Kac labels*, play also a role, in particular through their sums,

$$h = 1 + \sum_i a_i, \quad h^\vee = 1 + \sum_i a_i^\vee. \quad (3.37)$$

The numbers h and $h^\vee$ are respectively the *Coxeter number* and the *dual Coxeter number*. When a normalisation of roots has to be picked, which we have not done yet, one usually imposes that $\langle \theta, \theta \rangle = 2$.

Lastly the diagonalisation of the symmetrized Cartan matrix

$$\hat{C}_{ij} := 2 \frac{\langle \alpha_i, \alpha_j \rangle}{\sqrt{\langle \alpha_i, \alpha_i \rangle \langle \alpha_j, \alpha_j \rangle}} \quad (3.38)$$

yields a spectrum of eigenvalues

$$\text{eigenvalues of } \hat{C} = \left\{ 4 \sin^2 \left(\frac{\pi}{2h} m_i \right) \right\}, \quad i = 1, \dots, \ell, \quad (3.39)$$

in which a new set of integers m_i appears, the *Coxeter exponents*, satisfying $1 \leq m_i \leq h - 1$ with possible multiplicities. These numbers are relevant for various reasons. They contain useful information on the Weyl group. After addition of 1, (making them ≥ 2), one gets the degrees of algebraically independent Casimir operators, or the degrees where the Lie group has a non trivial cohomology, etc etc.

Examples: for A_{n-1} alias $\mathfrak{su}(n)$, roots and coroots coincide. The highest root is $\theta = \sum_i \alpha_i$, thus $h = h^\vee = n$, the Coxeter exponents are $1, 2, \dots, n - 1$. For D_n alias $\mathfrak{so}(2n)$, roots and coroots are again identical, $\theta = \alpha_1 + 2\alpha_2 + \dots + 2\alpha_{n-2} + \alpha_{n-1} + \alpha_n$, $h = 2n - 2$, and the exponents are $1, 3, \dots, 2n - 3, n - 1$, with $n - 1$ double if n is even.

See Appendix F for Tables of data on the classical simple algebras.

3.3 Representations of semi-simple algebras

3.3.1 Weights. Weight lattice

We now turn our attention to representations of semi-simple algebras, with an approach parallel to that of previous sections. In what follows, “representation” means finite dimensional irreducible representation. We also assume these representations to be unitary: this is the case of interest for representations of compact groups. The elements of the Cartan subalgebra commute among themselves, they also commute in any representation. Denoting with “bras” and “kets” the vectors of that representation, and writing simply X (instead of $d(X)$) for the representative of the element $X \in \mathfrak{g}$, one may find a basis $|\lambda_a\rangle$ which diagonalises simultaneously the elements of the Cartan algebra

$$H|\lambda_a\rangle = \lambda(H)|\lambda_a\rangle \quad (3.40)$$

or equivalently

$$H_i|\lambda_a\rangle = \lambda_{(i)}|\lambda_a\rangle, \quad (3.41)$$

with an eigenvalue λ which is again a linear form on the space $\mathfrak{h}$, hence an element of $\mathfrak{h}^*$, the root space. Such a vector $\lambda = (\lambda_{(i)})$ of $\mathfrak{h}^*$ is called a *weight*. Note that for a unitary representation, the H are Hermitian, hence λ is real-valued: the weights are real vectors of $\mathfrak{h}^*$. As the eigenvalue λ may occur with some multiplicity, we have appended the eigenvectors with a multiplicity index a . The set of weights of a given representation forms in the space $\mathfrak{h}^*$ the *weight diagram* of the representation, see Fig. 3.5 below for examples in the case of $\mathfrak{su}(3)$.

The adjoint representation is a particular representation of the algebra whose weights are the roots. The roots studied in the previous sections thus belong to the set of weights in $\mathfrak{h}^*$.

The vectors $|\lambda_a\rangle$ forming a basis of the representation, their total number, including the multiplicity, equals the dimension of the representation space E . This space E contains representation subspaces for each of the $\mathfrak{su}(2)$ algebras that we identified in § 3.2, generated by $\{H_\alpha, E_\alpha, E_{-\alpha}\}$. By the same argument as in § 3.2, we shall now show that any weight λ satisfies

$$\forall \alpha, \quad 2 \frac{\langle \lambda, \alpha \rangle}{\langle \alpha, \alpha \rangle} = m' \in \mathbb{Z}, \quad (3.42)$$

and conversely, it may be shown that any $\lambda \in \mathfrak{h}^*$ satisfying (3.42) is the weight of some finite dimensional representation. One may thus use (3.42) as an alternative definition of weights. To convince oneself that the weights of any representation satisfy (3.42), one may, like in § 3.2, define the maximal chain of weights through λ

$$\lambda + p'\alpha, \dots, \lambda, \dots, \lambda + q'\alpha \quad p' \leq 0, \quad q' \geq 0,$$

which form a representation of the $\mathfrak{su}(2)$ subalgebra, and then show that $m' = -p' - q'$.

Let p' be the smallest ≤ 0 integer such that $(E_{-\alpha})^{|p'|}|\lambda_a\rangle \neq 0$, and q' the largest ≥ 0 integer such that $(E_\alpha)^{q'}|\lambda_a\rangle \neq 0$, H_α has respective eigenvalues $\langle \lambda, \alpha \rangle + p'\langle \alpha, \alpha \rangle$, and $\langle \lambda, \alpha \rangle + q'\langle \alpha, \alpha \rangle$ on these vectors. Expressing that the eigenvalues of $2H_\alpha/\langle \alpha, \alpha \rangle$ are opposite integers, one finds

$$2q' + 2 \frac{\langle \lambda, \alpha \rangle}{\langle \alpha, \alpha \rangle} = 2j \quad 2p' + 2 \frac{\langle \lambda, \alpha \rangle}{\langle \alpha, \alpha \rangle} = -2j. \quad (3.43)$$

Subtracting these equations gives $q' - p' = 2j$, and the length of the chain is $2j + 1$ (dimension of the spin j representation of $\mathfrak{su}(2)$), while adding them to get rid of $2j$, one has

$$2 \frac{\langle \lambda, \alpha \rangle}{\langle \alpha, \alpha \rangle} = -(q' + p') =: m', \quad \text{as announced in (3.42).}$$

This chain is invariant under the action of the Weyl reflection w_α . (This is a generalisation of the $\mathbb{Z}_2$ symmetry of $\mathfrak{su}(2)$ “multiplets” $(-j, -j + 1, \dots, j - 1, j)$.) More generally the set of weights is invariant under the Weyl group: if λ is a weight of a representation, so is $w_\alpha(\lambda)$, and one shows that they have the same multiplicity. The weight diagram of a representation is thus invariant under the action of W .

The set of weights is split by the Weyl group W into “chambers”, whose number equals the order of W . The chamber associated with the element w of W is the cone

$$\mathcal{C}_w = \{ \lambda | \langle w\lambda, \alpha_i \rangle \geq 0, \forall i = 1, \dots, \ell \}, \quad (3.44)$$

where the α_i are the simple roots. (This is not quite a partition, as some weights belong to the “walls” between chambers.) The *fundamental chamber* is $\mathcal{C}_1$, corresponding to the identity in W . The weights belonging to that fundamental chamber are called *dominant weights*. Any weight may be brought into $\mathcal{C}_1$ by some operation of W : it is on the “orbit” (for the Weyl group) of a unique dominant weight. Among the weights of a representation, at least one belongs to $\mathcal{C}_1$.

On the other hand, from $[H_i, E_\alpha] = \alpha_{(i)} E_\alpha$ follows that

$$H_i E_\alpha |\lambda_a\rangle = ([H_i, E_\alpha] + E_\alpha H_i) |\lambda_a\rangle = (\alpha_{(i)} + \lambda_{(i)}) E_\alpha |\lambda_a\rangle$$

hence that $E_\alpha |\lambda_a\rangle$, if non vanishing, is an eigenvector of weight $\lambda + \alpha$. Now, in an irreducible representation, all vectors are obtained from one another by such actions of E_α , and we conclude that

▷ *Two weights of the same (irreducible) representation differ by a integer-coefficient combination of roots,*

(but this combination is in general not a root).

One then introduces a partial order on weights of the same representation: $\lambda' > \lambda$ if $\lambda' - \lambda = \sum_i n_i \alpha_i$, with non negative (integer) coefficients n_i . Among the weights of that representation, one proves there exists a unique highest weight Λ , which is shown to be of multiplicity 1. The highest weight vector will be denoted $|\Lambda\rangle$ (with no index a). It is such that for any positive root $E_\alpha |\Lambda\rangle = 0$, (otherwise, it would not be the highest), hence $q' = 0$ in equation (3.43) and $\langle \Lambda, \alpha \rangle = \frac{1}{2} \langle \alpha, \alpha \rangle j > 0$, Λ is thus a dominant weight.

▷ *The highest weight of a representation is a dominant weight, $\Lambda \in \mathcal{C}_1$.*

This highest weight vector characterises the irreducible representation. (In the case of $\mathfrak{su}(2)$, this would be a vector $|j, m = j\rangle$.) In other words, two representations are equivalent iff they have the same highest weight.

One then introduces the *Dynkin labels* of the weight λ by

$$\lambda_i = 2 \frac{\langle \lambda, \alpha_i \rangle}{\langle \alpha_i, \alpha_i \rangle} \in \mathbb{Z} \quad (3.45)$$

with α_i the simple roots. For a dominant weight, thus for any highest weight of a representation, these indices are non negative, *i.e.* in $\mathbb{N}$.

The *fundamental weights* Λ_i satisfy by definition

$$2 \frac{\langle \Lambda_j, \alpha_i \rangle}{\langle \alpha_i, \alpha_i \rangle} = \delta_{ij} . \quad (3.46)$$

Their number equals the rank ℓ of the algebra, and they make a basis of $\mathfrak{h}^*$. Each one is the highest weight of an irreducible representation called *fundamental*; hence there are ℓ fundamental representations. We have thus obtained

▷ Any irreducible representation is characterised by its highest weight,

and with a little abuse of notation, we denote (Λ) the irreducible representation of highest weight Λ .

▷ Any highest weight decomposes on fundamental weights, and its components are its Dynkin labels (3.45),

$$\Lambda = \sum_{j=1}^{\ell} \lambda_j \Lambda_j \quad , \quad \lambda_i \in \mathbb{N} . \quad (3.47)$$

and any Λ of the form (3.47) is the highest weight of an irreducible representation.

Stated differently, the knowledge of the fundamental weights suffices to construct all irreducible representations of the algebra.

Using the properties just stated, show that the highest weight of the adjoint representation is necessarily θ , defined in § 3.2.4.

Weight and root lattices

Generally speaking, given a basis of vectors $e_1, \dots, e_p$ in a p dimensional space, the *lattice* generated by these vectors is the set of vectors $\sum_{i=1}^p z^i e_i$ with coefficients $z^i \in \mathbb{Z}$. This lattice is also denoted $\mathbb{Z}e_1 + \dots + \mathbb{Z}e_p$.

The *weight lattice* P is the lattice generated by the ℓ fundamental weights Λ_i . The root lattice Q is the one generated by the ℓ simple roots α_i . This is a sublattice of P . Any weight of an irreducible representation belongs to P .

One may consider the *congruence classes* of the additive group P wrt its subgroup Q , that are the classes for the equivalence relation $\lambda \sim \lambda'$ iff $\lambda - \lambda' \in Q$. The number $|P/Q|$ of these classes turns out to be equal to the determinant of the Cartan matrix(!). In the case of $\mathfrak{su}(n)$, there are n classes, we shall return to that point later.

One may also introduce the lattice $Q^\vee$ generated by the ℓ coroots $\alpha_i^\vee$ (cf §2.4). It is the “dual” of P , in the sense that $\langle \alpha_i^\vee, \Lambda_j \rangle \in \mathbb{Z}$.

One also shows that the subgroups of the finite group P/Q are isomorphic to homotopy groups of groups G having $\mathfrak{g}$ as a Lie algebra! For example for $\mathfrak{su}(n)$, we find below that $P/Q = \mathbb{Z}_n$, and these subgroups are characterised by a divisor d of n . For each of them, $\mathrm{SU}(n)/\mathbb{Z}_d$ has the $\mathfrak{su}(n)$ Lie algebra. The case $n = 2$, with $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$, is quite familiar.

Dimension and Casimir operator

It may be useful to know the dimension of a representation with a given highest weight and the value of the quadratic Casimir operator in that representation. These expressions are given in terms of the *Weyl vector* ρ , defined by any of the two (non trivially!) equivalent formulas

$$\begin{aligned}\rho &= \frac{1}{2} \sum_{\alpha > 0} \alpha \\ &= \sum_j \Lambda_j .\end{aligned}\quad (3.48)$$

A remarkable formula, due to Weyl, gives the dimension of the representation of highest weight Λ as a product over positive roots

$$\dim(\Lambda) = \prod_{\alpha > 0} \frac{\langle \Lambda + \rho, \alpha \rangle}{\langle \rho, \alpha \rangle} \quad (3.49)$$

while the eigenvalue of the quadratic Casimir reads

$$C_2(\Lambda) = \frac{1}{2} \langle \Lambda, \Lambda + 2\rho \rangle . \quad (3.50)$$

A related question is that of the trace of generators of $\mathfrak{g}$ in the representation (Λ) . Let t_a be the basis of $\mathfrak{g}$ such that $\text{tr } t_a t_b = T_A \delta_{ab}$, with a coefficient T_A (“ A ” like adjoint) whose sign depends on conventions (t Hermitian or antihermitian, see Chap. 1). In the representation of highest weight Λ , one has

$$\text{tr } d_\Lambda(t_a) d_\Lambda(t_b) = T_\Lambda \delta_{ab} . \quad (3.51)$$

But in that basis, the quadratic Casimir reads $C_2 = \sum_a (d_\Lambda(t_a))^2$ hence, taking the trace,

$$\begin{aligned}\text{tr } C_2 &= \sum_a \text{tr } (d_\Lambda(t_a))^2 = T_\Lambda \sum_a 1 = T_\Lambda \dim \mathfrak{g} \\ &= C_2(\Lambda) \text{tr } I_\Lambda = C_2(\Lambda) \dim(\Lambda)\end{aligned}\quad (3.52)$$

whence

$$T_\Lambda = C_2(\Lambda) \frac{\dim(\Lambda)}{\dim \mathfrak{g}} , \quad (3.53)$$

a useful formula in calculations (gauge theories ...). In the adjoint representation, $\dim(A) = \dim \mathfrak{g}$, hence $T_\Lambda = T_A = C_2(A)$.

There is a host of additional, sometimes intriguing, formulas relating various aspects of Lie algebras and representation theory. For example the Freudenthal–de Vries “strange formula”, which connects the norms of the vectors ρ and θ to the dimension of the algebra and the Coxeter number: $\langle \rho, \rho \rangle = \frac{h}{24} \langle \theta, \theta \rangle \dim \mathfrak{g}$.

There is also a formula (Freudenthal) giving the multiplicity of a weight λ within a representation of given highest weight Λ . And last, as a related issue, a formula by Weyl giving the character $\chi_\Lambda(e^H)$ of that representation evaluated on an element of the *Cartan torus*, an abelian subgroup resulting from the exponentiation of the Cartan algebra $\mathfrak{h}$.

Conjugate representation

Given a representation of highest weight Λ , its complex conjugate representation is generally non equivalent. One may characterize what is its highest weight $\bar{\Lambda}$ thanks to the Weyl group. The non-equivalence of representations (Λ) and $(\bar{\Lambda})$ has to do with the symmetries of the Dynkin diagram. For the algebras of type B, C, E_7, E_8, F_4, G_2 for which there is no non trivial symmetry, the representations are self-conjugate. This is also the case of D_{2r} . For the others, conjugation corresponds to the following symmetry on Dynkin labels

$$\begin{array}{lll} A_\ell = \text{su}(\ell + 1) & \lambda_i \leftrightarrow \lambda_{\ell+1-i} & \ell > 1 \\ D_{2r+1} = \text{so}(4r + 2) & \lambda_\ell \leftrightarrow \lambda_{\ell-1} , & \ell = 2r + 1 \\ E_6 & \lambda_i \leftrightarrow \lambda_{6-i} , & i = 1, 2 . \end{array} \quad (3.54)$$

3.3.2 Roots and weights of $\mathfrak{su}(n)$

Let us construct explicitly the weights and thus the irreducible representations of $\mathfrak{su}(n)$.

We first pick a convenient parametrization of the space $\mathfrak{h}^*$, which is of dimension $n - 1$. Let e_i , $i = 1, \dots, n$, be n vectors of $\mathfrak{h}^* = \mathbb{R}^{n-1}$ (hence necessarily dependent), satisfying $\sum_1^n e_i = 0$. They are obtained starting from an orthonormal basis $\hat{e}_i$ of $\mathbb{R}^n$ by projecting the $\hat{e}_i$ on an hyperplane orthogonal to $\hat{\rho} := \sum_{i=1}^n \hat{e}_i$, thus $e_i = \hat{e}_i - \frac{1}{n}\hat{\rho}$. It is convenient to choose the hyperplane $\sum_i x^i = 1$ in the space $\mathbb{R}^n$. These vectors have scalar products given by

$$\langle e_i, e_j \rangle = \delta_{ij} - \frac{1}{n}. \quad (3.55)$$

In terms of these vectors, the positive roots of $\mathfrak{su}(n) = A_{n-1}$, whose number equals $|\Delta_+| = n(n-1)/2$, are

$$\alpha_{ij} = e_i - e_j, \quad 1 \leq i < j \leq n, \quad (3.56)$$

and the $\ell = n - 1$ simple roots are

$$\alpha_i = \alpha_{i,i+1} = e_i - e_{i+1}, \quad 1 \leq i \leq n - 1. \quad (3.57)$$

These roots have been normalized by $\langle \alpha, \alpha \rangle = 2$. The sum of positive roots is easily computed to be

$$\begin{aligned} 2\rho &= (n-1)e_1 + (n-3)e_2 + \dots + (n-2i+1)e_i + \dots - (n-1)e_n \\ &= (n-1)\alpha_1 + 2(n-2)\alpha_2 + \dots + i(n-i)\alpha_i + \dots + (n-1)\alpha_{n-1}. \end{aligned} \quad (3.58)$$

One checks that the Cartan matrix is

$$C_{ij} = \langle \alpha_i, \alpha_j \rangle = \begin{cases} 2 & \text{if } i = j \\ -1 & \text{if } i = j \pm 1 \end{cases}$$

in accord with the Dynkin diagram of type A_{n-1} , thus justifying (3.57). The fundamental weights Λ_i , $i = 1, \dots, n-1$ are then readily written

$$\Lambda_i = \sum_{j=1}^i e_j, \quad (3.59)$$

$$e_1 = \Lambda_1, \quad e_i = \Lambda_i - \Lambda_{i-1} \text{ for } i = 2, \dots, n-1, \quad e_n = -\Lambda_{n-1} \quad (3.60)$$

with scalar products

$$\langle \Lambda_i, \Lambda_j \rangle = \frac{i(n-j)}{n}, \quad i \leq j. \quad (3.61)$$

The Weyl group $W \equiv \mathcal{S}_N$ acts on roots and on weights by permuting the e_i :

$$w \in W \leftrightarrow \bar{w} \in \mathcal{S}_N : w(e_i) = e_{\bar{w}(i)}.$$

Dimension of the representation of weight Λ

Combining formulas (3.49) and (3.56), prove the following expression

$$\dim(\Lambda) = \prod_{1 \leq i < j \leq n} \frac{f_i - f_j + j - i}{j - i} \quad \text{où} \quad f_i := \sum_{k=i}^{n-1} \lambda_k, \quad f_n = 0. \quad (3.62)$$

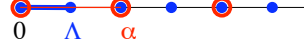


Figure 3.3: Weights of $\mathfrak{su}(2)$. The positive parts of the weight (small dots) and root (big dots) lattices.

Conjugate representations

If $\Lambda = (\lambda_1, \dots, \lambda_{n-1})$ is the highest weight of an irreducible representation of $\mathfrak{su}(n)$, $\bar{\Lambda} = (\lambda_{n-1}, \dots, \lambda_1)$ is that of the complex conjugate, generally inequivalent, representation (see above in § 3.3.1). Note that neither the dimension, nor the value of the quadratic Casimir operator distinguish the representations Λ and $\bar{\Lambda}$.

“ n -ality”.

There are n congruence classes of P with respect to Q . They are distinguished by the value of

$$\nu(\lambda) := \lambda_1 + 2\lambda_2 + \dots + (n-1)\lambda_{n-1} \pmod{n}, \quad (3.63)$$

to which one may give the unimaginative name of “ n -ality”, by extension of the “triality” of $\mathfrak{su}(3)$, see below. The elements of the root lattice thus have $\nu(\lambda) = 0$.

Examples of $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$.

In the case of $\mathfrak{su}(2)$, there is one fundamental weight $\Lambda = \Lambda_1$ and one positive root α , normalised by $\langle \alpha, \alpha \rangle = 2$, hence $\langle \Lambda, \alpha \rangle = 1$, $\langle \Lambda, \Lambda \rangle = \frac{1}{2}$. Thus $\alpha = 2\Lambda$, Λ corresponds to the spin $\frac{1}{2}$ representation, α to spin 1. The weight lattice and the root lattice are easy to draw, see Fig. 3.3. The Dynkin label λ_1 is identical to the integer $2j$, the two congruence classes of P wrt Q correspond to representations of integer and half-integer spin, the dimension $\dim(\Lambda) = \lambda_1 + 1 = 2j + 1$ and the Casimir operator $C_2(\Lambda) = \frac{1}{4}\lambda_1(\lambda_1 + 2) = j(j + 1)$, in accord with well known expressions.

For $\mathfrak{su}(3)$, the weight lattice is triangular, see Fig. (3.4), on which the triality $\tau(\lambda) := \lambda_1 + 2\lambda_2 \pmod{3}$ has been shown and the fundamental weights and the highest weights of the “low lying” representations have been displayed. Following the common use, representations are referred to by their dimension⁵

$$\dim(\Lambda) = \frac{1}{2}(\lambda_1 + 1)(\lambda_2 + 1)(\lambda_1 + \lambda_2 + 2), \quad (3.64)$$

supplemented by a bar to distinguish a representation from its conjugate, whenever necessary. The conjugate of representation of highest weight $\Lambda = (\lambda_1, \lambda_2)$ has highest weight $\bar{\Lambda} = (\lambda_2, \lambda_1)$. Only the representations lying on the bisector of the Weyl chamber are thus self-conjugate.

Exercise. Compute the eigenvalue of the quadratic Casimir operator in terms of Dynkin labels λ_1, λ_2 using the formulas (3.50) and (3.58).

⁵which may be ambiguous; for example, identify on Fig. (3.4) the weight of another representation of dimension 15.

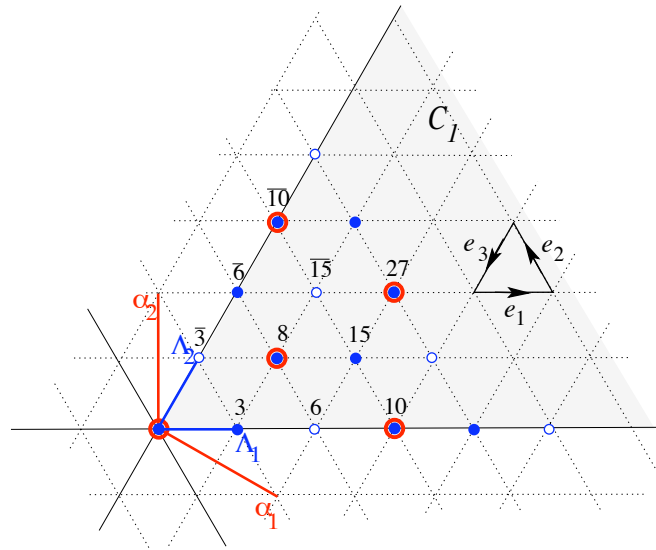


Figure 3.4: Weights of $\mathfrak{su}(3)$. Only the first Weyl chamber $\mathcal{C}_I$ has been detailed, with some highest weights. The weights of triality 0 (forming the root lattice) are represented by a wide disk, those of triality 1, resp. 2, by a full, resp. open disk.

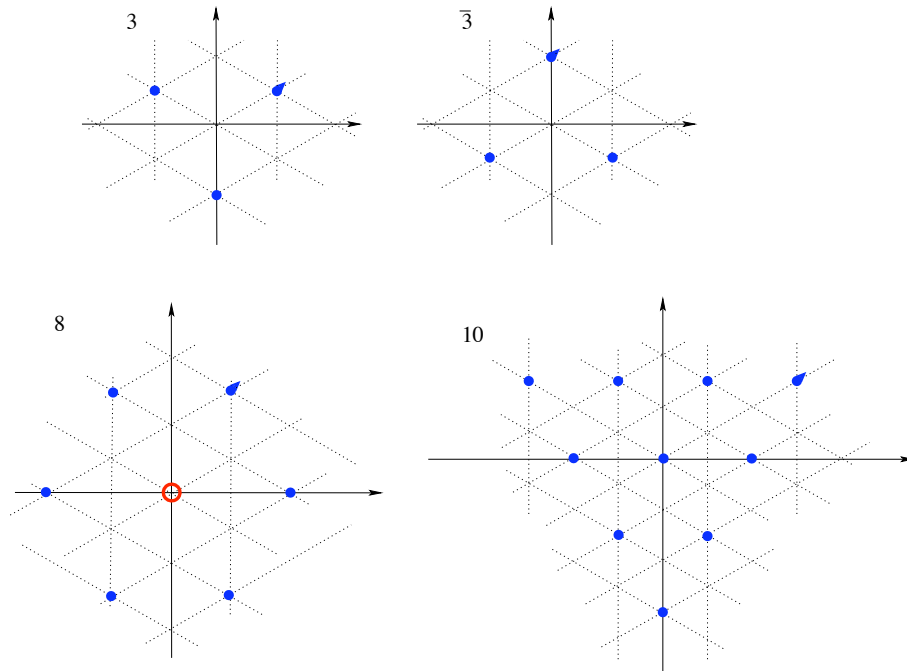


Figure 3.5: The weight diagrams of low lying representations of $\mathfrak{su}(3)$, denoted by their dimension. Note that a rotation of 30° of the weight lattice has been performed with respect to the previous figure. In each representation, the highest weight is marked by a small indentation. The small dots are weights of multiplicity 1, the wider and open dots have multiplicity 2.

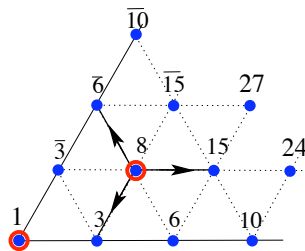


Figure 3.6: Tensor product of the 8 representation by the 3 representation, depicted on the weight diagram of $\mathfrak{su}(3)$.

The set of weights of low lying representations is displayed on Fig. (3.5), after a rotation of the axes of the previous figures. The horizontal axis, colinear to α_1 , and the vertical axis, colinear to Λ_2 , will indeed acquire a physical meaning: that of axes of isospin and “hypercharge” coordinates, see next chapter.

Remark. The case of $\mathfrak{su}(n)$ has been detailed. Analogous formulas for roots, fundamental weights, etc of other simple algebras are of course known explicitly and tabulated in the literature. See for example Appendix F for the identity card of “classical algebras”, of type A, B, C, D , and Bourbaki, chap.6, for more details on the other algebras.

3.4 Tensor products of representations of $\mathfrak{su}(n)$

3.4.1 Littlewood-Richardson rules

Given two irreducible representations of $\mathfrak{su}(n)$ (or of any other Lie algebra), a frequently encountered problem is to decompose their tensor product into a direct sum of irreducible representations. If one is only interested in multiplicities and if one has a character table of the corresponding compact *group*, one may use the formulae proved in Chap. 2, § 2.3.2.

There exists also fairly complex rules giving that decomposition into irreducible representations of a product of two irreducible representations (Λ) and (Λ') of $\mathfrak{su}(n)$. Those are the Littlewood-Richardson rules, which appeal to the expression in terms of Young tableaux (see next §). But it is often simpler to proceed step by step, noticing that the irreducible representation (Λ') is found in an adequate product of fundamental representations, and examining the successive products of representation Λ by these fundamental representations. By the associativity of the tensor product, one brings the original problem back to that of the tensor product of (Λ) by the various fundamental representations.

The latter operation is easy to describe on the weight lattice. Given the highest weight Λ in the first Weyl chamber $\mathcal{C}_1$, the tensor product of (Λ) by the fundamental representation of highest weight Λ_i is decomposed into irreducible representations in the following way: one adds in all possible ways the $\dim(\Lambda_i)$ weights of the fundamental to the vector Λ and one keeps as highest weights in the decomposition only the weights resulting from this addition that belong to $\mathcal{C}_1$.

Let us illustrate that on the case of $\mathfrak{su}(3)$. Suppose that we want to determine the decomposition of $8 \otimes 8$. One knows that the 8 representation (adjoint) is to be found in the product of two fundamental 3 and $\bar{3}$. The weights of the fundamental representation “3” of highest weight $\Lambda_1 = e_1$ are e_1, e_2, e_3 . Those of the fundamental representation “ $\bar{3}$ ” are their opposites. With the previous rule, one finds

$$\begin{aligned} 3 \otimes 3 &= \bar{3} + 6 & 3 \otimes \bar{3} &= 1 + 8 \\ 3 \otimes 6 &= 8 + 10 & 3 \otimes \bar{6} &= \bar{3} + \bar{15} \\ 3 \otimes 8 &= 3 + \bar{6} + 15 \\ 3 \otimes 15 &= 6 + \bar{15} + 24 \end{aligned} \tag{3.65}$$

etc, and their conjugates, see Fig. 3.6. In general one adds the three vectors $e_1 = (1, 0)$, $e_2 = (-1, 1)$ et $e_3 = (0, -1)$ (in the basis Λ_1, Λ_2) to $\Lambda = (\lambda_1, \lambda_2)$: the highest weights of the decomposition are thus $(\lambda_1 + 1, \lambda_2)$, $(\lambda_1 - 1, \lambda_2 + 1)$ and $(\lambda_1, \lambda_2 - 1)$, among which those having a negative Dynkin label are discarded. Note the consistency with triality: all the representations appearing in the rhs have the same triality, the sum (modulo 3) of trialities of those of the lhs. For example, $\tau(3) = 1$, $\tau(15) = 1$, $\tau(6) = 2$, $\tau(\bar{15}) = 2$, etc.

Iterating this procedure, one then computes

$$8 \otimes (1 + 8) = 8 \otimes 3 \otimes \bar{3} = (3 + \bar{6} + 15) \otimes \bar{3} = 1 + 8 + 8 + 8 + 10 + \bar{10} + 27$$

from which one derives the formula

$$8 \otimes 8 = 1 + 8_s + 8_a + 10 + \bar{10} + 27 . \tag{3.66}$$

In the latter expression, one added a subscript s or a to distinguish the two copies of the 8 representation: one is symmetric, the other antisymmetric in the exchange of the two representations 8 of the left hand side. This relation will be very useful in the following chapter, in the study of the $SU(3)$ “flavor” symmetry group.

Though a bit tedious, this procedure is simple and systematic. There exists a more elaborate graphical rule for the tensor product of two general highest weight representations (Λ) and (Λ') . There exist also codes computing these decompositions...

3.4.2 Explicit tensor construction of representations of $SU(2)$ and $SU(3)$

Consider a vector $V \in \mathbb{C}^n$ in the defining representation of $SU(n)$. Under the action of $U \in SU(n)$, $V \mapsto V' = UV$, or component-wise $v^i \mapsto v'^i = U^i_j v^j$, with indices $i, j = 1, \dots, n$. Let W be a vector which transforms by the complex conjugate representation, (like $W = V^*$), hence $W \mapsto W' = U^* W$. It is natural to denote the components of W with lower indices, since $U^* = (U^\dagger)^T$, and therefore $w'_i = w_j (U^\dagger)^j_i$. Note that $V \cdot W := v^i w_i$ is invariant, by virtue of $U^\dagger U = I$. In other words the mixed tensor δ^i_j is invariant

$$\delta'^i_j = U^i_{i'} U^{\dagger j'}_j \delta^{i'}_{j'} = (U U^\dagger)^i_j = \delta^i_j .$$

Consider now tensors of rank (p, m) , with p upper indices and m lower ones, transforming as $V^{\otimes p} \otimes W^{\otimes m}$, hence according to

$$t'^{i_1 \dots i_p}_{k_1 \dots k_m} = U^{i_1}_{j_1} \dots U^{i_p}_{j_p} U^{\dagger l_1}_{k_1} \dots U^{\dagger l_m}_{k_m} t^{j_1 \dots j_p}_{l_1 \dots l_m}. \quad (3.67)$$

• In the case of $SU(2)$, we know that the representations U and U^* are equivalent. This results from the existence of a matrix $C = i\sigma_2$, such that $CUC^{-1} = U^*$, thus CV^* transforms like V . Or, since $C_{ij} = \epsilon_{ij}$ and $\epsilon_{i'j'} U^{i'}_i U^{j'}_j = \epsilon_{ij} \det U = \epsilon_{ij}$, the antisymmetric tensor ϵ , invariant and invertible ($\epsilon^{ij} = -\epsilon_{ij}$, $\epsilon_{ij} \epsilon^{jk} = \delta_i^k$), may be used to raise or lower indices, ($v_i := \epsilon_{ij} v^j$, hence $v_1 = v^2$, $v_2 = -v^1$); and therefore it suffices to consider only tensors of rank p with upper indices. For any pair of indices, say i_1 and i_2 , such a tensor may be written as a sum of symmetric and antisymmetric components in these indices

$$t^{i_1 i_2 \dots i_p} = t^{[i_1, i_2] \dots i_p} + t^{\{i_1, i_2\} \dots i_p}$$

with $t^{[i_1, i_2] \dots i_p} := \frac{1}{2}(t^{i_1 i_2 \dots i_p} - t^{i_2 i_1 \dots i_p})$ and $t^{\{i_1, i_2\} \dots i_p} := \frac{1}{2}(t^{i_1 i_2 \dots i_p} + t^{i_2 i_1 \dots i_p})$. The antisymmetric component may be recast as $t^{[i_1, i_2] \dots i_p} = \epsilon^{i_1 i_2} \tilde{t}^{i_3 \dots i_p}$, with $\tilde{t}^{i_3 \dots i_p} = -\frac{1}{2} \epsilon_{ab} t^{abi_3 \dots i_p}$, and its rank has thus been reduced⁶. Consequently only tensors that are completely symmetric in all their p upper indices give irreducible representations, and one recovers once again the construction of all irreducible representations of $SU(2)$ by symmetrized tensor products of the representation of dimension 2, see Chap. 0, and the rank p identifies with $2j$. One checks in particular that the number of independent components of a rank p completely symmetric tensor in the space $\mathbb{C}^2$ is $p + 1$, since these components have $0, 1, \dots, p$ indices equal to 1, the other being equal to 2.

A rank p completely symmetric tensor will be represented by a “Young diagram” with p boxes . For the general definition of a Young diagram, see next section. Take $p = 3$ for definiteness. The tensor product of such a rank 3 tensor by a rank 1 tensor will be depicted as

$$\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \otimes \square = \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array}$$

which means, in terms of components,

$$4t^{ijk}u^l = (t^{ijk}u^l + t^{jkl}u^i + t^{ikl}u^j + t^{ijl}u^k) + (t^{ijk}u^l - t^{jkl}u^i) + (t^{ijk}u^l - t^{ikl}u^j) + (t^{ijk}u^l - t^{ijl}u^k)$$

where the first term is completely symmetric in its $p + 1 = 4$ indices, and the following terms are antisymmetric in (i, l) , (j, l) or (k, l) . According to the previous argument, the latter may be reduced to rank 2 tensors.

$$(t^{ijk}u^l - t^{jkl}u^i) = \epsilon^{il} \tilde{t}^{jk}, \quad \tilde{t}^{jk} = -\epsilon_{ab} t^{ajk}u^b$$

which we represent by erasing the columns with two boxes. Therefore

$$\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \otimes \square = \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array}$$

⁶It may be useful to recall the identities $\epsilon_{ab}\epsilon_{cd} = \delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc}$ and hence $\epsilon_{ab}\epsilon_{bc} = -\delta_{ac}$.

where we recognize the familiar rule $j \otimes \frac{1}{2} = (j + \frac{1}{2}) \oplus (j - \frac{1}{2})$.

Exercise : reproduce with this method the decomposition rule of $j \otimes j'$.

• In the case of $SU(n)$, $n > 2$, one must consider tensors with two types of indices, upper and lower, and reduce them. But this is only in the case of $SU(3)$ that this construction will provide us with all irreducible representations. For $n > 3$ one has to introduce other tensors transforming under fundamental representations of $SU(n)$ other than the defining representation (of dimension n) and its conjugate.

We thus restrict the discussion of the end of this section to the case of $SU(3)$. The tensors are of type $t_{j_1 \dots j_m}^{i_1 \dots i_p}$, $(i, j = 1, 2, 3)$, transforming under the representation $3^{\otimes p} \otimes \bar{3}^{\otimes m}$. We still have an invariant tensor ϵ , but now of rank 3,

$$\epsilon_{i'j'k'} U^{i'}_i U^{j'}_j U^{k'}_k = \epsilon_{ijk} \det U = \epsilon_{ijk} ,$$

which allows us to trade any pair of upper antisymmetric indices for a lower one, or vice versa, and thus to reduce the rank. But a pair of one upper and one lower indices may also be contracted, according to a remark at the beginning of the section. Consequently one may consider only completely symmetric and traceless tensors of rank (p, m) . One may prove that such tensors form an irreducible representation, which is nothing other than the representation of highest weight $p\Lambda_1 + m\Lambda_2$ in the notations of § 3.3.2. We content ourselves with a check that the dimensions of these representations are in accord with those given in (3.64), see Exercise E. With this representation we associate again a Young diagram with two rows, the first with $p + m$, the second with m boxes.

The rules of decomposition of tensor products, in particular by the fundamental representations 3 and $\bar{3}$ (see § 3.4.1), are also recovered in this language : the new box must be added in all possible ways to the diagram (while preserving the decreasing of lengths of rows), and any column of height 3 is erased, reflecting the property that $\det U = 1$. Exercise : study the reduction of $\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & & \\ \hline \end{array} \otimes \square$ and recover the graphical rule of § 3.4.1 in this language.

A particular case that we shall use repeatedly in the next chapter is the following: the adjoint representation is that of rank (1,1) traceless tensors. This is no surprise: the adjoint representation is spanned by the $\mathfrak{su}(3)$ Lie algebra, hence by (anti)Hermitian 3×3 traceless matrices. A tensor of that representation transforms by $t_j^i \mapsto t'^i_j = U^i_{i'} U^{*j'}_j t'^{i'}_{j'}$, or in a matrix form

$$t' = U t U^\dagger , \quad (3.68)$$

which is also expected, compare with the definition of the adjoint representation in Chap. 2. Which Young diagram is associated with the adjoint representation?

3.5 Young tableaux and representations of $GL(n)$ and $SU(n)$

The previous construction extends to $\mathfrak{su}(n)$, in fact to the group $GL(n)$, and involves symmetrization and anti-symmetrization operations related to the symmetric group of permutations S_m . We just give a few indications.

Let $E = \mathbb{C}^n$ be the vector space of dimension n . The group $\text{GL}(n)$ is naturally represented in E

$$g \in \text{GL}(E) \simeq \text{GL}(n), \quad x \in E \mapsto x' = g.x . \quad (3.69)$$

Form the m -th tensor power of E : $F = E^{\otimes m} = E \otimes \cdots \otimes E$. In F , the group $\text{GL}(n)$ acts by a representation, the m -th tensor power of (3.69)

$$g \in \text{GL}(n), \quad D(g)x^{(1)} \otimes \cdots \otimes x^{(m)} = g.x^{(1)} \otimes \cdots \otimes g.x^{(m)} \quad (3.70)$$

which is in general reducible. But in F , there is also the action of the symmetric group S_m according to

$$\sigma \in S_m, \quad \widehat{D}(\sigma)x^{(1)} \otimes \cdots \otimes x^{(m)} = x^{(\sigma^{-1}1)} \otimes \cdots \otimes x^{(\sigma^{-1}m)} . \quad (3.71)$$

Choose a basis e_i in E , and denote g_{ij} the matrix elements of $g \in \text{GL}(n)$ in that basis. The representation of $\text{GL}(n)$ in F has a matrix

$$\mathcal{D}(g)_{\{i_1 \cdots i_m\} \{j_1 \cdots j_m\}} = \prod_{k=1}^m g_{i_k j_k} \quad (3.72)$$

and that of S_m

$$\widehat{\mathcal{D}}(\sigma)_{\{i_1 \cdots i_m\} \{j_1 \cdots j_m\}} = \prod_{k=1}^m \delta_{i_{\sigma k} j_k} . \quad (3.73)$$

A tensor t , element of F , has components $t^{i \cdot}$ in that basis and transforms under the action of $g \in \text{GL}(n)$, resp. of $\sigma \in S_m$, into a tensor t' of components $t'^{i \cdot} = \mathcal{D}_{i \cdot, j \cdot} t^{j \cdot}$, resp. $\widehat{\mathcal{D}}_{i \cdot, j \cdot} t^{j \cdot}$. These two sets of matrices commute

$$\begin{aligned} \sum_{\{j \cdot\}} \mathcal{D}(g)_{\{i \cdot\}, \{j \cdot\}} \widehat{\mathcal{D}}(\sigma)_{\{j \cdot\}, \{k \cdot\}} &= \prod_l g_{i_l j_l} \delta_{j_l, k_{\sigma^{-1}l}} = \prod_l g_{i_{\sigma l} k_l} \\ &= \sum_{\{j \cdot\}} \widehat{\mathcal{D}}(\sigma)_{\{i \cdot\}, \{j \cdot\}} \mathcal{D}(g)_{\{j \cdot\}, \{k \cdot\}} . \end{aligned} \quad (3.74)$$

Define then a *Young diagram*. A Young diagram is made of m boxes set in k rows of non increasing length: $f_1 \geq f_2 \geq \cdots \geq f_k$, $\sum f_i = m$. Here is an example for $m = 8$, with $f_1 = 4, f_2 = 2, f_3 = 2$



The m boxes of a Young diagram may then be filled with different integers ranging between 1 and m , thus making a *Young tableau*. A *standard tableau* is a tableau in which the integers are increasing in each row from left to right, and in each column from top to bottom.

The number n_Y of standard tableaux obtained from a Young diagram Y is computed as follows. One defines the numbers $\ell_i = f_i + k - i$, $i = 1, \cdots, k$. They form a strictly increasing sequence: $\ell_1 < \ell_2 < \cdots < \ell_k$. Then one proves that

$$n_Y = \frac{n!}{\prod_i \ell_i!} \prod_{i < j} (\ell_i - \ell_j) \quad (3.75)$$

where the product in the numerator is 1 if there is a single row.

The representation theory of the symmetric group S_m tells us that there is a bijection between irreducible representations and Young diagrams with m boxes. The dimension of that representation is given by the number of standard tableaux (3.75).

A tensor is said to be of (symmetry) type Y if it transforms by S_m under that representation. The commutation of matrices $\mathcal{D}(g)$ and $\widehat{\mathcal{D}}(\sigma)$ then ensures that tensors of type Y form an invariant sub-space under the action of $\text{GL}(n)$.

Example. Consider the cases of $m = 2$ and $m = 3$. In the first case, rank 2 tensors may be decomposed into their symmetric and antisymmetric parts which transform independently under the action of $\text{GL}(n)$

$$t^{i_1 i_2} = \frac{1}{2} (t^{i_1 i_2} + t^{i_2 i_1}) + \frac{1}{2} (t^{i_1 i_2} - t^{i_2 i_1}) .$$

This decomposition corresponds to the two Young tableaux with 2 boxes, arranged horizontally or vertically. For rank 3, one writes in a similar way the tensors associated with the 4 standard Young tableaux

$$\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline \end{array} \quad A = t^{i_1 i_2 i_3} + t^{i_2 i_3 i_1} + t^{i_3 i_1 i_2} + t^{i_2 i_1 i_3} + t^{i_3 i_2 i_1} + t^{i_1 i_3 i_2} \quad (3.76)$$

$$\begin{array}{|c|} \hline 1 \\ \hline 2 \\ \hline 3 \\ \hline \end{array} \quad B = t^{i_1 i_2 i_3} + t^{i_2 i_3 i_1} + t^{i_3 i_1 i_2} - t^{i_2 i_1 i_3} - t^{i_3 i_2 i_1} - t^{i_1 i_3 i_2} \quad (3.77)$$

$$\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array} \quad C_1 = t^{i_1 i_2 i_3} - t^{i_2 i_3 i_1} + t^{i_2 i_1 i_3} - t^{i_3 i_2 i_1} \quad (3.78)$$

$$\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array} \quad D_1 = t^{i_1 i_2 i_3} - t^{i_2 i_1 i_3} + t^{i_3 i_2 i_1} - t^{i_3 i_1 i_2} \quad (3.79)$$

where, to make the notations lighter, the indices i_1, i_2, i_3 on $A, \dots, D_1$ have been omitted. Any rank 3 tensor decomposes on that basis:

$$6t^{i_1 i_2 i_3} = A + B + 2(C_1 + D_1).$$

The labels 1 on C and D recall that under the action of the group S_3 , these objects mix with another combination $C_2 = t^{i_1 i_3 i_2} - t^{i_3 i_1 i_2} + t^{i_2 i_3 i_1} - t^{i_1 i_2 i_3}$, (resp. $D_2 = t^{i_2 i_1 i_3} + t^{i_2 i_3 i_1} - t^{i_1 i_2 i_3} - t^{i_1 i_3 i_2}$) of t^{ijk} to make dimension 2 representations. On the contrary the action of the group $GL(n)$ mixes the different components of tensor A , those of tensor B , etc. Tensors C et D transform by equivalent representations.

All Young tableaux, however, do not contribute for a given n . It is clear that a tableau with $k > n$ rows implies an antisymmetrization of k indices taking their values in $\{1, \dots, n\}$ and vanishes. On the other hand it is easy to see that any tableau with $k \leq n$ rows gives rise to a representation. One proves, and we admit, that this representation of $GL(n)$ is irreducible and that its dimension is

$$\dim_Y^{(n)} = \frac{\Delta(f_1 + n - 1, f_2 + n - 2, \dots, f_n)}{\Delta(n - 1, n - 2, \dots, 0)} \quad (3.80)$$

where $\Delta(a_1, a_2, \dots, a_n) = \prod_{i < j} (a_i - a_j)$ is the *Vandermonde determinant* of the a 's and the f_i denote the lengths of rows of the tableau Y . This is a polynomial of degree $m = \sum f_i$ in n . Compare with (3.62).

In the case of a one-row tableau, the formula results from a simple combinatorial argument. The dimension equals the number of components of the completely symmetric tensor $t^{i_1 \dots i_m}$ in which one may assume that $1 \leq i_1 \leq i_2 \leq \dots \leq i_m \leq n$. One has to arrange in all possible ways $n - 1 <$ signs between the m indices $i_1, \dots, i_m$ to mark the successive blocks de 1, 2, ..., n . The seeked dimension is thus the binomial coefficient $\binom{n+m-1}{n} \equiv C_{n+m-1}^n$, in accord in this particular case with (3.80).

In the previous example with $m = 3$, the last two tensors C_1 and D_1 transform according to equivalent representations. One thus says that $E^{\otimes 3}$ decomposes as

$$\begin{array}{|c|c|c|} \hline & & \\ \hline \end{array} + \begin{array}{|c|} \hline \\ \hline \end{array} + 2 \begin{array}{|c|c|} \hline & \\ \hline \end{array}$$

where the third representation comes with a multiplicity two. As a general rule, the multiplicity in $E^{\otimes m}$ of some representation of $GL(n)$ labeled by a Young tableau equals the dimension of the corresponding representation of S_m .

This remarkable relation between representations of S_m and of $GL(n)$ is due to Frobenius and H. Weyl and is called *Frobenius–Weyl duality*.

One may extend these considerations to other groups of linear transformations, $SL(n)$, $O(n)$, $U(n)$, ... Because of the additional conditions on the g matrices in these groups, a further reduction of the representations may occur. For example, we saw in sect. 2.2.2 that the tensor power $E^{\otimes 2}$ of the 3-dimensional Euclidian space reduced under the action of $SO(3)$ into three subspaces, corresponding to tensors with a definite symmetry and traceless, and to an invariant scalar.

Relations between Young diagrams and weights of $\mathfrak{su}(n)$

Let us finally give the relation between the two descriptions of irreducible representations obtained for $SU(n)$ or its Lie algebra $\mathfrak{su}(n)$. In that case, one may limit the number of rows of the Young tableau Y to $k \leq n - 1$

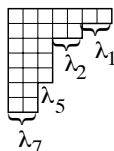


Figure 3.7: Correspondence between a Young diagram and a highest weight (or Dynkin labels). Here $Y \leftrightarrow \Lambda = (2, 2, 0, 0, 1, 0, 2)$

to obtain all irreducible representations. The i -th fundamental weight is represented by a Young diagram made of one column of height i , for example $\Lambda_3 = \begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix}$. And the correspondence between the highest weight Λ with Dynkin labels λ_i and the tableau Y with rows of length f_i is as follows

$$\Lambda = (\lambda_1, \dots, \lambda_{n-1}) \leftrightarrow Y = (f_i = \sum_{j=i}^{n-1} \lambda_j) . \quad (3.81)$$

In other words, λ_k is the number of columns of Y of height k , see Fig. 3.7.

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A short bibliography (cont'd)

The construction of roots and weights is described in many references given above: (Bump; Bröcker et Dieck; Gilmore ...) but also in J. E. Humphreys, *Introduction to Lie Algebras and Representation Theory*, Graduate Texts in Mathematics 9, Springer.

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On octonions and exceptional groups, look at the exhaustive article by John C. Baez, *Bull. Amer. Math. Soc.* 39 (2002), 145-205 (also available on line); or P. Ramond, *Group Theory, A Physicist's Survey*, Cambridge 2010.

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Appendix F. The classical algebras of type A, B, C, D

F.1 $\mathfrak{sl}(N) = A_{N-1}$

Rank $= l = N - 1$, dimension $N^2 - 1$, Coxeter number $h = N$, dual Coxeter number $h^\vee = N$.

$e_i, i = 1, \dots, N$ a set of vectors in $\mathbb{R}^N$ such that $\sum_{i=1}^N e_i = 0, \langle e_i, e_j \rangle = \delta_{ij} - \frac{1}{N}$.

Roots $\alpha_{ij} = e_i - e_j, i \neq j = 1, \dots, N$; positive roots $\alpha_{ij} = e_i - e_j, i < j$; their number $|\Delta_+| = N(N-1)/2$; simple roots $\alpha_i := \alpha_{i+1} = e_i - e_{i+1} i = 1, \dots, N-1$.

Highest root $\theta = \alpha_1 + \dots + \alpha_{N-1} = 2e_1 + e_2 + \dots + e_{N-1} = \Lambda_1 + \Lambda_{N-1} = (1, 0, \dots, 0, 1)$.

Sum of positive roots

$$\begin{aligned} 2\rho &= (N-1)e_1 + (N-3)e_2 + \dots + (N-2i+1)e_i + \dots - (N-1)e_N \\ &= (N-1)\alpha_1 + 2(N-2)\alpha_2 + \dots + i(N-i)\alpha_i + \dots + (N-1)\alpha_{N-1}. \end{aligned} \quad (\text{F-1})$$

$$\text{Cartan matrix } \langle \alpha_i, \alpha_j \rangle = \begin{cases} 2 & \text{if } i = j \\ -1 & \text{if } i = j \pm 1 \\ 0 & \text{otherwise} \end{cases}$$

Fundamental weights $\Lambda_i i = 1, \dots, N-1, \Lambda_i = \sum_{j=1}^i e_j, e_1 = \Lambda_1, e_i = \Lambda_i - \Lambda_{i-1}$ for $i = 2, \dots, N-1, e_N = -\Lambda_{N-1}$.

$\langle \Lambda_i, \Lambda_j \rangle = \frac{i(N-j)}{N}$ for $i \leq j$.

Weyl group: $W \equiv \mathcal{S}_N$ acts on the weights by permuting the e_i : $w \in W \leftrightarrow \bar{w} \in \mathcal{S}_N$: $w(e_i) = e_{\bar{w}(i)}$

Coxeter exponents $\{1, 2, \dots, N-1\}$.

F.2 $\mathfrak{so}(2l+1) = B_l, l \geq 2$

Rank $= l$, dimension $l(2l+1)$, Coxeter number $h = 2l$, dual Coxeter number $h^\vee = 2l-1$

$e_i, i = 1, \dots, l, \langle e_i, e_j \rangle = \delta_{ij}$ a basis of $\mathbb{R}^l$.

Roots $\pm e_i, 1 \leq i \leq l$ and $\pm e_i \pm e_j, 1 \leq i < j \leq l$. Basis of simple roots $\alpha_i = e_i - e_{i+1}, i = 1, \dots, l-1$, and $\alpha_l = e_l$.

Positive roots

$$\begin{aligned} e_i &= \sum_{i \leq k \leq l} \alpha_k, \quad 1 \leq i \leq l, \\ e_i - e_j &= \sum_{i \leq k < j} \alpha_k, \quad 1 \leq i < j \leq l, \\ e_i + e_j &= \sum_{i \leq k < j} \alpha_k + 2 \sum_{j \leq k \leq l} \alpha_k, \quad 1 \leq i < j \leq l, \end{aligned} \quad (\text{F-2})$$

their number is $|\Delta_+| = l^2$.

Highest root $\theta = e_1 + e_2 = \alpha_1 + 2\alpha_2 + \dots + 2\alpha_l$.

Sum of positive roots

$$\begin{aligned} 2\rho &= (2l-1)e_1 + (2l-3)e_2 + \dots + (2l-2i+1)e_i + \dots + 3e_{l-1} + e_l \\ &= (2l-1)\alpha_1 + 2(2l-2)\alpha_2 + \dots + i(2l-i)\alpha_i + \dots + l^2\alpha_l. \end{aligned} \quad (\text{F-3})$$

$$\text{Cartan matrix } \langle \alpha_i, \alpha_j^\vee \rangle = \begin{cases} 2 & \text{if } 1 \leq i = j \leq l \\ -1 & \text{if } 1 \leq i = (j \pm 1) \leq l-1 \\ -2 & \text{if } i = l-1, j = l \\ -1 & \text{if } i = l, j = l-1 \\ 0 & \text{otherwise} \end{cases}$$

Fundamental weights $\Lambda_i = \sum_{j=1}^i e_j$, $i = 1, \dots, l-1$, $\Lambda_l = \frac{1}{2} \sum_{j=1}^l e_j$; hence $e_1 = \Lambda_1 = (1, 0, \dots, 0)$, $e_i = \Lambda_i - \Lambda_{i-1} = (0, \dots, -1, 1, 0, \dots)$, $i = 2, \dots, l-1$, $e_l = 2\Lambda_l - \Lambda_{l-1} = (0, \dots, 0, -1, 2)$.

Dynkin labels of the roots

$\alpha_1 = (2, -1, 0, \dots)$, $\alpha_i = (0, \dots, -1, 2, -1, 0, \dots)$, $i = 2, \dots, l-2$; $\alpha_{l-1} = (0, \dots, 0, -1, 2, -2)$; $\alpha_l = (0, \dots, 0, -1, 2)$ and $\theta = (0, 1, 0, \dots, 0)$

Weyl group: $W \equiv \mathcal{S}_l \ltimes (\mathbb{Z}_2)^l$, of order $2^l \cdot l!$, acts on the weights by permuting the e_i and $e_i \mapsto (\pm 1)_i e_i$.

Coxeter exponents $\{1, 3, 5, \dots, 2l-1\}$.

F.3. $\mathfrak{sp}(2l) = C_l$, $l \geq 2$

Rank = l , dimension $l(2l+1)$, Coxeter number $h = 2l$, dual Coxeter number $h^\vee = l+1$

e_i , $i = 1, \dots, l$, $\langle e_i, e_j \rangle = \frac{1}{2} \delta_{ij}$ a basis of $\mathbb{R}^l$ (**Beware !** factor 2 to enforce the normalisation $\theta^2 = 2$). Basis of simple roots $\alpha_i = e_i - e_{i+1}$, $i = 1, \dots, l-1$, and $\alpha_l = 2e_l$.

Roots $\pm 2e_i$, $1 \leq i \leq l$ and $\pm e_i \pm e_j$, $1 \leq i < j \leq l$.

Positive roots

$$\begin{aligned} e_i - e_j &= \sum_{i \leq k < j} \alpha_k, \quad 1 \leq i < j \leq l, \\ e_i + e_j &= \sum_{i \leq k < j} \alpha_k + 2 \sum_{j \leq k < l} \alpha_k + \alpha_l, \quad 1 \leq i < j \leq l, \\ 2e_i &= 2 \sum_{i \leq k < l} \alpha_k + \alpha_l, \quad 1 \leq i \leq l, \end{aligned} \tag{F-4}$$

their number is $|\Delta_+| = l^2$.

Highest root $\theta = 2e_1 = 2\alpha_1 + 2\alpha_2 + \dots + 2\alpha_{l-1} + \alpha_l$.

Sum of positive roots

$$\begin{aligned} 2\rho &= 2e_1 + (2l-2)e_2 + \dots + (2l-2i+2)e_i + \dots + 4e_{l-1} + 2e_l \\ &= 2l\alpha_1 + 2(2l-1)\alpha_2 + \dots + i(2l-i+1)\alpha_i + \dots + (l-1)(l-2)\alpha_{l-1} + \frac{1}{2}l(l+1)\alpha_l. \end{aligned} \tag{F-5}$$

$$\text{Cartan matrix } \langle \alpha_i, \alpha_j^\vee \rangle = \begin{cases} 2 & \text{if } 1 \leq i = j \leq l \\ -1 & \text{if } 1 \leq i = (j \pm 1) \leq l-1 \\ -1 & \text{if } i = l-1, j = l \\ -2 & \text{if } i = l, j = l-1 \\ 0 & \text{otherwise} \end{cases}$$

Fundamental weights $\Lambda_i = \sum_{j=1}^i e_j$, $i = 1, \dots, l$, hence $e_1 = \Lambda_1 = (1, 0, \dots, 0)$, $e_i = \Lambda_i - \Lambda_{i-1} = (0, \dots, -1, 1, 0, \dots)$, $i = 2, \dots, l$.

Dynkin labels of the roots

$\alpha_1 = (2, -1, 0, \dots)$, $\alpha_i = (0, \dots, -1, 2, -1, 0, \dots)$, $i = 2, \dots, l-1$; $\alpha_l = (0, \dots, 0, -2, 2)$ and $\theta = (2, 0, \dots, 0)$

Weyl group: $W \equiv \mathcal{S}_l \ltimes (\mathbb{Z}_2)^l$, of order $2^l \cdot l!$, acts on the weights by permuting the e_i and $e_i \mapsto (\pm 1)_i e_i$.

F.4. $\mathfrak{so}(2l) = D_l$, $l \geq 3$

Rank = l , dimension $l(2l-1)$, Coxeter number = dual Coxeter number $h = 2l-2 = h^\vee$

e_i , $i = 1, \dots, l$, $\langle e_i, e_j \rangle = \delta_{ij}$ a basis of $\mathbb{R}^l$.

Basis of simple roots $\alpha_i = e_i - e_{i+1}$, $i = 1, \dots, l-1$, and $\alpha_l = e_{l-1} + e_l$.

Positive roots

$$\begin{aligned}
e_i - e_j &= \sum_{i \leq k < j} \alpha_k, \quad 1 \leq i < j \leq l, \\
e_i + e_j &= \sum_{i \leq k < j} \alpha_k + 2 \sum_{j \leq k < l-1} \alpha_k + \alpha_{l-1} + \alpha_l, \quad 1 \leq i < j \leq l-1, \\
e_i + e_l &= \sum_{i \leq k \leq l-2} \alpha_k + \alpha_l, \quad 1 \leq i \leq l-1,
\end{aligned} \tag{F-6}$$

their number is $|\Delta_+| = l(l-1)$.

Highest root $\theta = e_1 + e_2 = \alpha_1 + 2\alpha_2 + \cdots + 2\alpha_{l-2} + \alpha_{l-1} + \alpha_l$.

Sum of positive roots

$$\begin{aligned}
2\rho &= 2(l-1)e_1 + 2(l-2)e_2 + \cdots + 2e_{l-1} \\
&= 2(l-1)\alpha_1 + 2(2l-3)\alpha_2 + \cdots + i(2l-i-1)\alpha_i + \cdots + \frac{l(l-1)}{2}(\alpha_{l-1} + \alpha_l).
\end{aligned} \tag{F-7}$$

Weyl group: $W \equiv \mathcal{S}_l \ltimes (\mathbb{Z}_2)^{l-1}$, of order $2^{l-1} \cdot l!$, acts on the weights by permuting the e_i and $e_i \mapsto (\pm 1)_i e_i$, with $\prod_i (\pm 1)_i = 1$.

Coxeter exponents $\{1, 3, 5, \dots, 2l-3, l-1\}$, with $l-1$ appearing twice if l is even.

$$\text{Cartan matrix } \langle \alpha_i, \alpha_j \rangle = \begin{cases} 2 & \text{if } 1 \leq i = j \leq l \\ -1 & \text{if } 1 \leq i = (j \pm 1) \leq l-2 \\ -1 & \text{if } (i, j) = (l-2, l) \text{ or } (l, l-2) \\ 0 & \text{otherwise} \end{cases}$$

Fundamental weights $\Lambda_i = \sum_{j=1}^i e_j = \alpha_1 + 2\alpha_2 + \cdots + (i-1)\alpha_{i-1} + i(\alpha_i + \cdots + \alpha_{l-2}) + \frac{i}{2}(\alpha_{l-1} + \alpha_l)$ for $i = 1, \dots, l-2$; $\Lambda_{l-1} = \frac{1}{2}(e_1 + \cdots + e_{l-1} - e_l) = \frac{1}{2}(\alpha_1 + 2\alpha_2 + \cdots + (l-2)\alpha_{l-2}) + \frac{l}{2}\alpha_{l-1} + \frac{l-2}{2}\alpha_l$; $\Lambda_l = \frac{1}{2}(e_1 + \cdots + e_{l-1} + e_l) = \frac{1}{2}(\alpha_1 + 2\alpha_2 + \cdots + (l-2)\alpha_{l-2}) + \frac{l-2}{2}\alpha_{l-1} + \frac{l}{2}\alpha_l$.

For the exceptional algebras of types E, F, G , see Bourbaki.

Exercises for chapter 3

A. Cartan algebra and roots

1. Show that any element X of $\mathfrak{g}$ may be written as $X = \sum x^i H_i + \sum_{\alpha \in \Delta} x^\alpha E_\alpha$ with the notations of § 3.1.2.

For an arbitrary H in the Cartan algebra, determine the action of $\text{ad}(H)$ on such a vector X ; conclude that $\text{ad}(H)\text{ad}(H')X = \sum_{\alpha \in \Delta} x^\alpha \alpha(H)\alpha(H')E_\alpha$ and taking into account that the eigenspace of each root α has dimension 1, cf point (*) 2. of § 3.1.2, that the Killing form reads

$$(H, H') = \text{tr}(\text{ad}(H)\text{ad}(H')) = \sum_{\alpha \in \Delta} \alpha(H)\alpha(H') . \quad (3-81)$$

2. One wants to show that roots α defined by (3.5) or (3.6) generate all the dual space $\mathfrak{h}^*$ of the Cartan subalgebra $\mathfrak{h}$. Prove that if it were not so, there would exist an element H of $\mathfrak{h}$ such that

$$\forall \alpha \in \Delta \quad \alpha(H) = 0 . \quad (3-82)$$

Using (3-81) show that this would imply $\forall H' \in \mathfrak{h}, (H, H') = 0$. Why is that impossible in a semi-simple algebra? (see the discussion before equation (3.10)).

3. Variant of the previous argument: under the assumption of 2. and thus of (3-82), show that H would commute with all H_i and all the E_α , thus would belong to the center of $\mathfrak{g}$. Prove that the center of an algebra is an abelian ideal. Conclude in the case of a semi-simple algebra.

B. Computation of the $N_{\alpha\beta}$

1. Show that the real constants $N_{\alpha\beta}$ satisfy $N_{\alpha\beta} = -N_{\beta\alpha}$ and, by complex conjugation of $[E_\alpha, E_\beta] = N_{\alpha\beta}E_{\alpha+\beta}$ that

$$N_{\alpha\beta} = -N_{-\alpha, -\beta} . \quad (3-83)$$

2. Consider three roots satisfying $\alpha + \beta + \gamma = 0$. Writing the Jacobi identity for the triplet $E_\alpha, E_\beta, E_\gamma$, show that $\alpha_{(i)}N_{\beta\gamma} + \text{cycl.} = 0$. Derive from it the relation

$$N_{\alpha\beta} = N_{\beta, -\alpha-\beta} = N_{-\alpha-\beta, \alpha} . \quad (3-84)$$

3. Considering the α -chain through β and the two integers p et q defined in § 3.2.1, write the Jacobi identity for $E_\alpha, E_{-\alpha}$ and $E_{\beta+k\alpha}$, with $p \leq k \leq q$, and show that it implies

$$\langle \alpha, \beta + k\alpha \rangle = N_{-\alpha, \beta+k\alpha}N_{\alpha, \beta+(k-1)\alpha} + N_{\beta+k\alpha, \alpha}N_{-\alpha, \beta+(k+1)\alpha} .$$

Let $f(k) := N_{\alpha, \beta+k\alpha}N_{-\alpha, -\beta-k\alpha}$. Using the relations (3-84), show that the previous equation may be recast as

$$\langle \alpha, \beta + k\alpha \rangle = f(k) - f(k-1) . \quad (3-85)$$

4. What are $f(q)$ et $f(q-1)$? Show that the recursion relation (3-85) is solved by

$$f(k) = -(N_{\alpha, \beta+k\alpha})^2 = (k-q)\langle \alpha, \beta + \frac{1}{2}(k+q+1)\alpha \rangle . \quad (3-86)$$

What is $f(p-1)$? Show that the expression (3.21) is recovered. Show that (3-86) is in accord with (3.23). The sign of the square root is still to be determined ..., see [Gi].

C. Study of the $B_l = so(2l + 1)$ and G_2 algebras1. $so(2l + 1) = B_l$, $l \geq 2$

- What is the dimension of the group $SO(2l + 1)$ or of its Lie algebra $so(2l + 1)$?
- What is the rank of the algebra? (Hint: diagonalize a matrix of $so(2l + 1)$ on $\mathbb{C}$, or write it as a diagonal of real 2×2 blocks, see lecture notes, §3.1)
- How many roots does the algebra have? How many positive roots? How many simple?
- Let e_i , $i = 1, \dots, l$ be a orthonormal basis in $\mathbb{R}^l$, $\langle e_i, e_j \rangle = \delta_{ij}$. Consider the set of vectors

$$\Delta = \{\pm e_i, 1 \leq i \leq l\} \cup \{\pm e_i \pm e_j, 1 \leq i < j \leq l\}$$

What is the cardinal of Δ ? Δ is the set of roots of the algebra $so(2l + 1)$.

- A basis of simple roots is given $\alpha_i = e_i - e_{i+1}$, $i = 1, \dots, l - 1$, et $\alpha_l = e_l$. Explain why the roots

$$\begin{aligned} e_i &= \sum_{i \leq k \leq l} \alpha_k, \quad 1 \leq i \leq l, \\ e_i - e_j &= \sum_{i \leq k < j} \alpha_k, \quad 1 \leq i < j \leq l, \\ e_i + e_j &= \sum_{i \leq k < j} \alpha_k + 2 \sum_{j \leq k \leq l} \alpha_k, \quad 1 \leq i < j \leq l, \end{aligned} \tag{3-87}$$

qualify as positive roots. Check that assertion on the case of $B_2 = so(5)$.

- Compute the Cartan matrix and check that it agrees with the Dynkin diagram given in the notes.

g. Compute the sum ρ of positive roots.

h. The Weyl is the (“semi-direct”) product $W \equiv \mathcal{S}_l \ltimes (\mathbb{Z}_2)^l$, which acts on the e_i (hence on weights and roots) by permutation and by independant sign changes $e_i \mapsto (\pm 1)_i e_i$. What is its order? In the case of B_2 , check that assertion and draw the first Weyl chamber.

i. Show that the vectors $\Lambda_i = \sum_{j=1}^i e_j$, $i = 1, \dots, l - 1$, $\Lambda_l = \frac{1}{2} \sum_{j=1}^l e_j$ are the fundamental weights.

j. Using Weyl formula: $\dim(\Lambda) = \prod_{\alpha > 0} \frac{\langle \Lambda + \rho, \alpha \rangle}{\langle \rho, \alpha \rangle}$ compute the dimension of the two fundamental representations of B_2 and of that of highest weight $2\Lambda_2$. In view of these dimensions, what are these representations of $SO(5)$?

k. Draw on the same figure the roots and the low lying weights of $so(5)$.

2. G_2

In the space $\mathbb{R}^2$, we consider three vectors e_1, e_2, e_3 of vanishing sum, $\langle e_i, e_j \rangle = \delta_{ij} - \frac{1}{3}$, and construct the 12 vectors

$$\pm(e_1 - e_2), \pm(e_1 - e_3), \pm(e_2 - e_3), \pm(2e_1 - e_2 - e_3), \pm(2e_2 - e_1 - e_3), \pm(2e_3 - e_1 - e_2)$$

They make the root system of G_2 , as we shall check.

- What can be said on the dimension of the algebra G_2 ?

- b. Show that $\alpha_1 = e_1 - e_2$ and $\alpha_2 = -2e_1 + e_2 + e_3$ are two simple roots, in accord with the Dynkin diagram of G_2 given in the notes. Compute the Cartan matrix.
- c. What are the positive roots? Compute the vector ρ , half-sum of positive roots.
- d. What is the group of invariance of the root diagram? Show that it is of order 12 and that it is the Weyl group of G_2 . Draw the first Weyl chamber.
- e. Check that the fundamental weights are

$$\Lambda_1 = 2\alpha_1 + \alpha_2 \quad \Lambda_2 = 3\alpha_1 + 2\alpha_2$$

- f. What are the dimensions of the fundament representations?
- g. In the two cases of B_2 and G_2 , one observes that the highest weight of the adjoint representation is given by the highest root. Explain why this is true in general.

3. A little touch of particle physics

Why were the groups $SO(5)$ or G_2 inappropriate as symmetry groups extending the $SU(2)$ isospin group, knowing that several “octets” of particles had been observed?

D. Root systems. Folding of Dynkin diagrams

One considers the simple roots α_i of the algebra $\mathfrak{su}(2n)$, numbered as in the lectures. (Beware! we do say $2n$!)

- 1. What is the rank of that algebra? What are the $\langle \alpha_i, \alpha_j \rangle$? Draw the corresponding Dynkin diagram. What is the symmetry of that diagram?
- 2. One then defines $\beta_i = (\alpha_i + \alpha_{2n-i})/\sqrt{2}$, for $i = 1, \dots, n-1$ and $\beta_n = \alpha_n/\sqrt{2}$. Calculate the $\langle \beta_i, \beta_j \rangle$.
- 3. Show that the β form a root system and identify it.
- 4. More generally, any system of simple roots with the same lengths may be “folded” according to a possible symmetry of its Dynkin diagram and then gives rise to another Dynkin diagram. *With no calculation*, which diagram should be obtained in that manner, starting from the E_6 diagram?

E. Dimensions of $SU(3)$ representations

We admit that the construction of § 3.4.2, of completely symmetric traceless rank (p, m) tensors in $\mathbb{C}^3$, does give the irreducible representations of $SU(3)$ of highest weight (p, m) . Then we want to determine the dimension $d(p, m)$ of the space of these tensors.

- 1. Show, by studying of the product of two tensors of rank $(p, 0)$ and $(0, m)$ and separating the trace terms (those containing a δ_j^i between one lower and one upper index) that $(p, 0) \otimes (0, m) = ((p-1, 0) \otimes (0, m-1)) \oplus (p, m)$ and thus that

$$d(p, m) = d(p, 0)d(0, m) - d(p-1)d(0, m-1).$$

- 2. Show by a computation analogous to that of $SU(2)$ that

$$d(p, 0) = d(0, p) = \frac{1}{2}(p+1)(p+2).$$

- 3. Derive from it the expression of $d(p, m)$ and compare with (3.64).